Semi-LM: A Reliable Multiperspective Consistency Semi-Supervised Learning Framework for Landslide Mapping From Remote Sensing Images

Shuhao Ran, *Graduate Student Member, IEEE*, Gang Ma, Jiangzhou Mei, Fudong Chi, Wei Zhou, and Yonghong Weng

***Abstract*— Deep learning techniques have achieved significant success in landslide mapping (LM) from remote sensing images. However, the need for large quantities of high-quality labels presents a substantial barrier to the practical application of deep learning models. Semi-supervised learning (SSL) offers a solution by leveraging the knowledge contained in unlabeled remote sensing landslide images (RSLIs) to improve model performance. The main challenges in applying SSL to LM include the limitations and appropriateness of consistency perturbation perspectives, as well as the potential disruption of both shallow and deep landslide features and microtopographic information. To address these challenges, we propose a reliable multiperspective consistency SSL framework called Semi-LM. This framework employs multiperspective consistency learning (MCL) as the learning pathway and introduces the dual-branch anchor CutMix (DACM) method and feature perturbation assign (FPA) strategy for LM. MCL achieves unified multiperspective perturbations by enforcing the alignment of multiperspective perturbed LM results with generated weak perturbation pseudo-labels, expanding the perturbation space and enabling model to learn more discriminative features from unlabeled RSLIs. DACM generates anchor masks by determining NO-Cut zones with landslide anchors, mitigating the risk of erroneous knowledge accumulation and improving the reliability of model optimization. Additionally, by measuring the local variability of the model within a high-dimensional feature space, FPA identifies the specific stage where low-density regions of separated classes are located. Extensive comparative and ablation experiments on five diverse landslide datasets demonstrated that Semi-LM significantly reduces the reliance on labels in LM, identifies more landslides, and provides more stable results.**

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I. Introduction

LANDSLIDES are natural disasters caused by the sliding failure of soil, rock, or artificial fill along slopes due to human, natural, or combined factors. They result in hundreds of deaths and billions of dollars in damage annually. With the rapid growth of the global population and the intensification of extreme weather events due to recent climate change, the risk of landslides is projected to increase sharply in the next decade [1]. Following extreme events, rapid and accurate landslide mapping (LM) is crucial for disaster relief, quantitative damage assessment, and post-disaster reconstruction [2].

The development of remote sensing technology has enhanced the use of remote sensing landslide images (RSLIs) for LM [3]. In recent years, deep learning has significantly advanced LM tasks by enabling automatic feature extraction and hierarchical representation learning. Integrating deep learning with remote sensing images for LM has become the mainstream approach. Based on the task type, deep learning-based LM can be classified into three categories: end-to-end semantic segmentation [4], [5], change detection [3], [6], and instance segmentation [7], [8].

Although deep learning-based methods have been widely applied to LM tasks, their success heavily relies on a large number of RSLIs and corresponding labels. While the advancement of satellite remote sensing technology makes acquiring vast amounts of RSLIs feasible, obtaining corresponding pixel-level annotations is extremely costly. This process typically requires substantial human resources and expertise in geography. As illustrated in Fig. 1, the detection capability of models for landslides improves significantly with the increase in labeled data in the training set, but this also substantially raises the cost of model training.

In this context, how to achieve more satisfactory results in deep learning-based LM with scarce labeled samples is a critical issue. Self-supervised learning and semi-supervised learning (SSL) have been proven to effectively enhance the performance of models by leveraging large amounts of unlabeled data. Therefore, exploring self-supervised learning and

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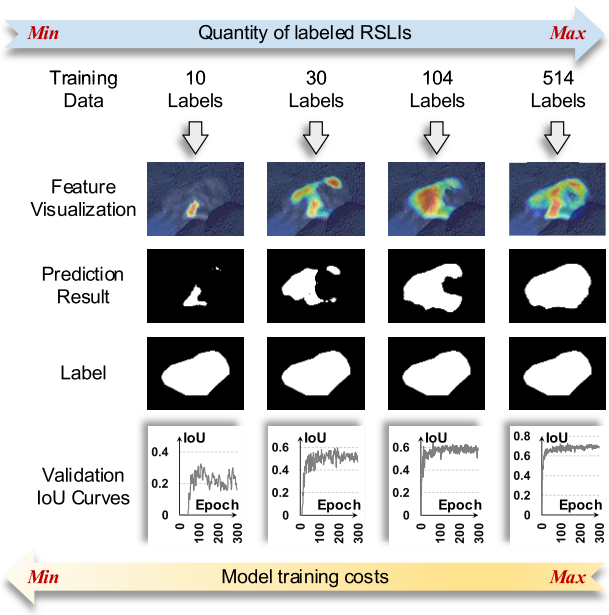


Fig. 1. Motivation of the SSL framework for LM. The addition of more labeled RSLIs enhances the precision of LM, yet concurrently raises the expenditure required for model training.

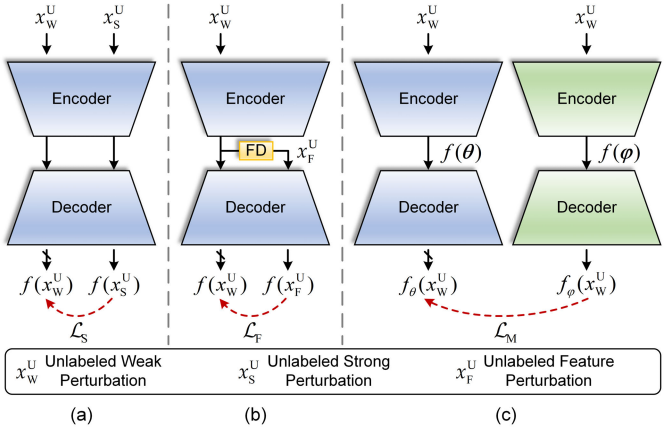


Fig. 2. Comparison between existing different perspective frameworks of consistency learning. (a) Input perspective. (b) Feature perspective. (c) Model perspective.

SSL to LM is one of the key approaches to addressing the shortage of labeled landslide samples.

The goal of self-supervised learning is to design pretext tasks that enable the model to learn general feature representations from large amounts of unlabeled data and fine-tune the model using a small amount of labeled data for a specific task. Currently, in the field of remote sensing, the most popular self-supervised learning methods include contrastive learning [9], [10], [11], [12] and masked image modeling (MIM) [13], [14], [15], [16]. Contrastive learning works by constructing positive and negative sample pairs, allowing the model to learn to pull similar samples closer and push dissimilar samples farther apart, thereby learning discriminative features. As one of the means to achieve this, data augmentation perturbations generate positive and negative sample pairs by applying spatial or spectral transformations to images, promoting knowledge learning from unlabeled remote sensing images [17]. For example, different augmented views of the same RSLI should exhibit more similar representations [18], which provides insights for learning robust feature representations from unlabeled RSLI data. Moreover, MIM trains the model to predict the contents of masked regions by learning contextual information, thus obtaining semantic representations of images [14]. In LM processing, such masking operations may obscure local and subtle topographic features of the landslide, making it difficult for the model to recover key information and potentially introducing false noise [19], leading to incomplete or erroneous semantic information in landslide images.

In contrast to the two-stage training process of self-supervised learning, SSL employs simultaneous learning on both unlabeled and labeled data. The SSL methods can be broadly categorized into generative adversarial networks (GANs), self-training, and consistency regularization. GANs, involving a generator and a discriminator, embed unlabeled data to create additional supervision signals for learning features [20], [21]. However, these methods are computationally complex and involve intricate training processes. The self-training uses a model pretrained on labeled data to infer labels for unlabeled data, incorporating these inferred labels as offline pseudo-labels into the labeled dataset for further training, thereby minimizing entropy [22], [23]. Nevertheless, self-training methods are susceptible to initial model errors, leading to confirmation bias issues. Consequently, current research often focuses on improving algorithms to obtain more accurate pseudo-labels. Consistency regularization has gained wide attention due to its simplicity and ease of integration into deep learning. Its core idea is that a robust model should maintain consistent results for the same input under perturbations, that is, the model output should not be affected by the perturbation. By constructing consistency regularization terms before and after perturbation, the model can learn general, shared features from unlabeled data, reducing overfitting to specific instance details and improving the generalization performance [24], [25].

Initially, SSL methods focused only on natural RGB images [26]. These methods did not account for the wide observation perspectives of remote sensing images, which include more diverse and complex interference from grounds. To address the challenges of SSL in remote sensing image segmentation, researchers have attempted to improve pseudo-label quality, introduce new data augmentation techniques, or develop new learning frameworks, and achieved some success. However, the feasibility of applying SSL with limited labeled data to LM has not been fully explored and validated.

Existing consistency regularization-based SSL methods face challenges when applied to LM, including limitations and appropriateness of consistency perturbation perspectives, as well as the potential disruption of both shallow and deep landslide features and microtopographic information. Specifically, as shown in Fig. 2, the consistency perturbation frameworks are limited to input perspectives, feature perspectives, and model perspectives, though effective, they hinder the

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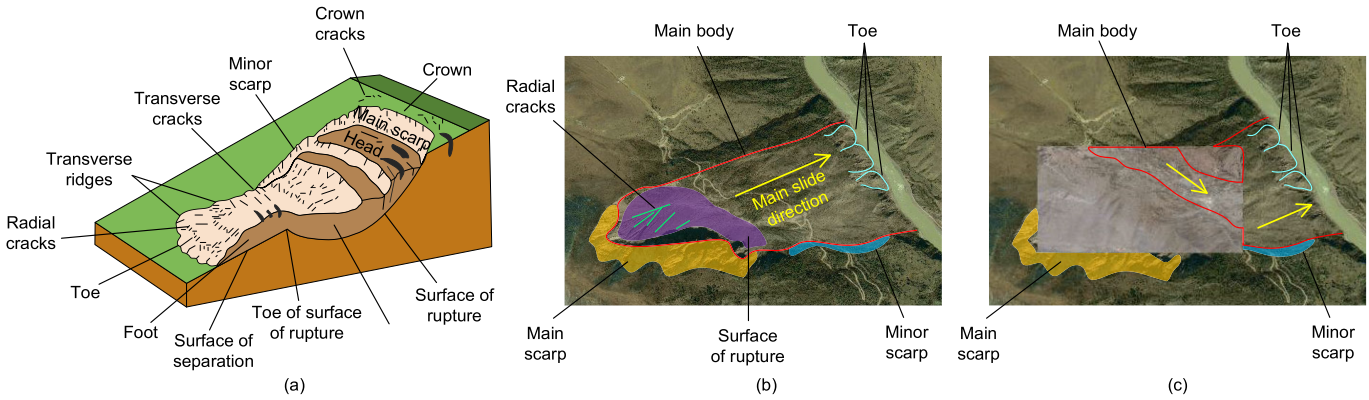


Fig. 3. Illustration of the disruption of landslide information by CutMix. (a) Main components of the landslide. (b) Partial landslide microtopographic in RSLI. (c) CutMix disrupted the spectral, textural continuity, semantic, and microtopographic information of landslides.

expansion of the perturbation space. This limitation restricts the model's ability to learn across multiperspective, ultimately resulting in insufficient utilization of the information contained in the limited unlabeled RSLIs. Landslides as geological disasters characterized by unique microtopographic features require the integration of spectral, texture, and topographic morphological features for effective RSLI-based LM, as illustrated in Fig. 3. Common strategies for weak-to-strong consistency, such as CutMix [27], involve randomly generated CutBox that may disrupt the spectral and textural continuity of landslides. Similar to self-supervised MIM, this disruption can damage the semantic and microtopographic information of landslides, leading to the learning and accumulation of erroneous features. Moreover, when introducing feature-perspective perturbations to overcome input perspective limitations, it is crucial to determine a feature perturbation assignment that aligns with the clustering assumption. This ensures the effectiveness of consistency regularization and avoids its failure. However, for RSLIs, the low-density regions separating classes do not correspond to the stages observed by [28] and [29], causing existing feature perturbation assignment strategies to fail in LM.

In response to the aforementioned problems, we propose a reliable multiperspective SSL framework for LM from remote sensing images, named Semi-LM. This framework incorporates multiperspective consistency learning (MCL) to expand and fully utilize the original perturbation space. MCL aligns input, feature, and model perturbation predictions into a unified objective function, enabling the learning of more discriminative features from unlabeled RSLIs. To enrich input perturbations while mitigating the disruption of original both shallow and deep features and microtopographic information of landslides, we design a dual-branch anchor CutMix (DACM) with landslide localization constraints. Similar to contrastive learning, DACM uses a common shared weak view to normalize two strong views, which can be interpreted as minimizing the distance between the strong views. This helps the model learn more discriminative feature representations. Additionally, considering the valuable information encoded in feature maps [29], we observe the clustering assumption in LM, determining the stage with the highest overlap between classification decision boundaries and low-density regions. Based on this observation, we propose the feature perturbation assign (FPA) strategy to optimize LM results.

In summary, the contributions of this article can be summarized as follows.

1) A reliable end-to-end multiperspective SSL framework for LM from remote sensing images is proposed. This framework integrates MCL into a unified objective function by aligning the landslide prediction results of input, feature, and model perturbations with weak perturbation pseudo-labels. This approach facilitates the efficient utilization of unlabeled RSLIs, significantly enhancing the accuracy of LM and achieving state-of-the-art (SOTA) performance on five landslide datasets.
2) We design a simple and effective DACM method. By defining a NO-Cut zone with a landslide anchor for the binary mask, DACM avoids the disruption of both shallow and deep landslide features and microtopographic information during dual-branch weak-to-strong consistency learning while providing richer information representations.
3) Through the calculation of local variability in the high-dimensional feature space of the model, we observe that the low-density regions separating classes are located at the end of the decoding stage. We propose the FPA strategy for LM, which fully exploits the valuable information encoded in feature maps to achieve the optimal LM performance.

II. Related Work

A. LM From Remote Sensing Images

In the past, LM was mainly conducted through visual interpretation of RSLIs. While this method yields accurate results, it is costly, labor-intensive, time-consuming, and relies heavily on human subjectivity [30], making it challenging to meet the needs of disaster prevention and mitigation. With the development of computer vision technology, computer-aided semiautomated and fully automated LM has rapidly developed and has been widely applied. Current computer-assisted methods for LM from remote sensing images can be divided into

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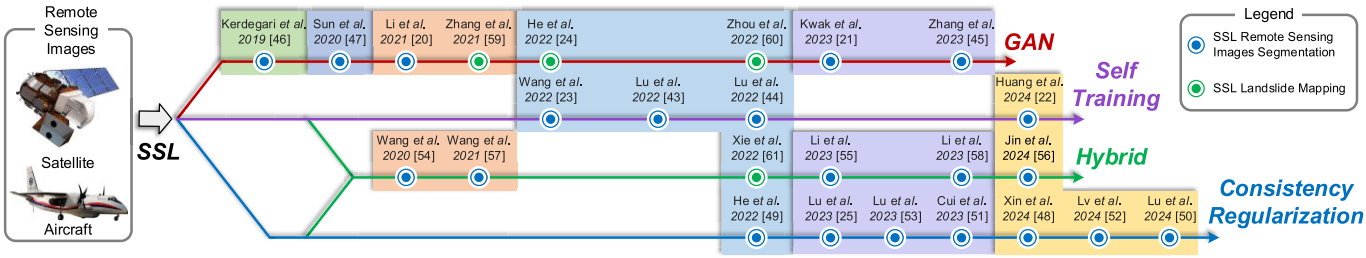


Fig. 4. Development of SSL for remote sensing image segmentation.

three categories: feature thresholding segmentation [5], [31], machine learning [32], [33], and deep learning [4], [34]. However, the feature thresholding segmentation method is prone to poor generalization due to the manual selection of feature thresholds and segmentation scales. Machine learning methods rely heavily on feature selection by specialists with specialized knowledge, limiting their efficiency and generalizability.

DCNN-based research on LM can be categorized into three types: end-to-end semantic segmentation, change detection, and instance segmentation. End-to-end semantic segmentation methods often use models such as ResUNet [5], U-Net [4], [35], DenseNett [34], and UNet++ [36]. Researchers have further improved LM accuracy by integrating multisource data [24], constructing multiscale feature extractors [37], or embedding attention mechanisms [38]. Due to the limitations of DCNNs in capturing global context, applying Transformers [39] to LM tasks has become a popular choice, such as ShapeFormer [40], CTDNet [41], and SCDUNet++ [2]. Additionally, LM based on change detection has also garnered attention [3], [6]. However, these methods have high data requirements and necessitate strict radiometric correction to accurately match image information [42]. The instance segmentation models not only accurately identify landslide locations but also evaluate the segmentation results of individual landslides [7], [8]. However, the complexity and computational demands of these methods may hinder their ability to meet the rapid LM needs post-earthquake.

B. Semi-Supervised Remote Sensing Image Segmentation

Consistent with SSL methods, semi-supervised remote sensing image segmentation can also be categorized into GANs, self-training, consistency regularization, and hybrid methods, as illustrated in Fig. 4. Self-training-based methods primarily focus on generating more accurate pseudo-labels to achieve better segmentation performance. Huang et al. [22] designed a decoupled weighting learning framework that separates the predictions of labeled and unlabeled data to reduce the impact of incorrect pseudo-labels on model training. To address the long-tailed distribution problem in SSL, APRST [43] and LSST [44] adopt pixel-level adaptive sampling to assign thresholds for different classes, thereby improving the quality of pseudo-labels. Similarly, ICNet [23] enhances pseudo-label quality by introducing a teacher model. GAN-based methods [20], [21], [45], [46], [47] generate pseudo-labels for unlabeled data using trained models and optimize the generation of pseudo-labels and model training with a discriminator. However, GANs inherently face challenges such as high computational costs and complex training processes. Consistency regularization-based methods introduce sparse dual-view cross-sample image generation [25], ClassMix [48], and semantic boundary information [49] to produce more diverse data augmentation results, thereby enriching input perturbation consistency. Moreover, introducing more diverse forms of perturbations [50], [51], [52], [53] enables consistency regularization to learn more representative feature representations. Hybrid methods [54], [55], [56] integrate multiple SSL strategies or other learning paradigms into the model training process, leveraging the strengths of both to better exploit the potential of unlabeled data. For example, RanPaste [57] combines consistency regularization with pseudo-label and uses thresholding to progressively improve the model segmentation performance. SegMind [58] integrates mean teacher (MT), MIM, and contrastive learning approaches. The MIM is used to enhance information interaction between different regions, while contrastive learning improves the linear separability and prediction confidence of the model.

Despite the partial success of semi-supervised remote sensing image segmentation, researchers focused on RSLIs segmentation for specific LM tasks remain scarce. SMDRF-Net [59] and He et al. [24] leverage GANs to mine multilayer representations from pseudo-labeled images, thereby learning discriminative features and generating pseudo-labels to reduce dependence on labeled data. Zhou et al. [60] build upon the GAN framework by incorporating a class activation map branch to determine the approximate location of landslides. In another notable study, Xie et al. [61] apply U^2PL [62] from natural image segmentation to LM in Luding County. This adaptation resulted in a significant improvement in segmentation accuracy, demonstrating the transferability and effectiveness of sophisticated SSL techniques in the LM domain. These studies suggest that SSL methods possess substantial untapped potential in the LM field, warranting further exploration and rigorous validation to fully exploit their capabilities.

C. Teacher–Student Framework

The teacher–student framework is a knowledge transfer paradigm that promotes the learning process through interaction and feedback, and it is widely used in SSL and self-supervised learning. During MT [63] training, the update of the student model using unlabeled images is entirely guided by the teacher model. The teacher model is updated via the

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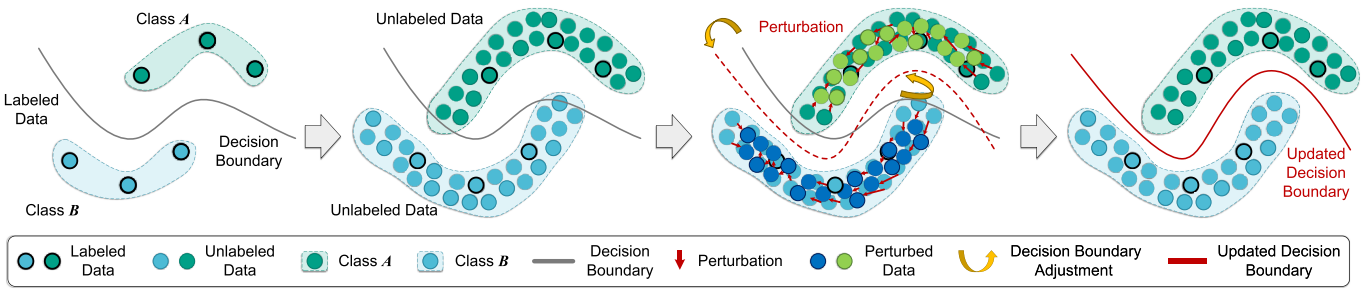


Fig. 5. Illustration of the process of consistency regularization using 2-D toy examples.

exponential moving average (EMA) of student model, ensuring consistency between the outputs of the teacher–student framework. In addition to its application in consistency regularization, the teacher–student framework can also be combined with pseudo-labeling. LSST [44] is a two-stage SSL method. First, the teacher model is trained using labeled remote sensing images, and then, the teacher model predicts the unlabeled images and outputs pseudo-labels to guide the update of the student model. Chen et al. [64] improve pseudo-label quality by adding dynamic threshold selection to the teacher model output and constructing the consistency self-training loss with the student model output. SegMind [58] combines the MT and self-supervised learning, introducing projector-r and projector-f into the student model to build multiple learning streams for updating parameters, achieving success in reducing demand on manual labels.

In self-supervised learning, the teacher–student framework is primarily used for knowledge distillation [65]. CMID [19] utilizes a teacher–student self-distillation architecture to learn representations from remote sensing images that are globally semantically separable and locally spatially perceptive. ObjSSL [66] combines self-distillation with a multiview pretext task to learn from unlabeled remote sensing images, encouraging the student model to learn the full spectral context from the teacher model. DK-SSCD [67] employs the MT network to improve the accuracy of self-supervised remote sensing change detection, effectively reducing the drastic fluctuations in accuracy during the inference process.

Normally, in both SSL and self-supervised learning, the teacher model plays the role of providing stable supervision or pseudo-labels, while the student model aligns feature representations to the teacher model. In the proposed Semi-LM, the hard pseudo-labels are provided by the student model, while the teacher model acts as the model perturbation. By enforcing consistency between teacher and student model outputs, models are motivated to learn more robust and generalizable features.

III. Methodology

To address the challenges faced in applying SSL to LM, this section first defines the problem of SSL in LM and consistency regularization. It then introduces the proposed SSL framework, Semi-LM, and elaborates on the DACM method and the details of the FPA strategy.

A. Problem Defined and Consistency Regularization

The shortage of adequate and usable labeled RSLIs remains a persistent problem in LM tasks. The goal of SSL-based LM is to train a model to perform per-pixel classification, under a small set of RSLI-label pairs dataset $S_{\mathrm{L}} = \{(x_i^{\mathrm{L}}, y_i^{\mathrm{L}})\}_{i=1}^{M_{\mathrm{L}}}$ and unlabeled RSLIs dataset $S_{\mathrm{U}} = \{(x_i^{\mathrm{U}})\}_{i=1}^{M_{\mathrm{U}}}$, where $M_{\mathrm{L}} \ll M_{\mathrm{U}}$. x_i represents the ith inputted RSLI. y_i^{L} denotes the manual label corresponding to the RSLI. Typically, the overall optimization objective in SSL-based LM tasks can be formulated as follows:

$$\mathcal{L} = \mathcal{L}_{\mathrm{Sup}} + \lambda_{\mathrm{U}} \cdot \mathcal{L}_{\mathrm{U}} \tag{1}$$

where $\mathcal{L}_{\mathrm{Sup}}$ is the supervised loss, which is calculated on the labeled dataset, and $\mathcal{L}_{\mathrm{U}}$ denotes the unsupervised loss, which is calculated on unlabeled dataset through different perspective consistency learning streams. λ_{U} represents the balance weight hyperparameter of unsupervised loss.

The ability of consistency learning terms to learn knowledge from unlabeled RSLIs is attributed to the theory of consistency regularization. Fig. 5 illustrates the consistency regularization process using 2-D toy examples. When only labeled data are available, it is possible to obtain decision boundaries with ease. However, after the addition of a large number of unlabeled 2-D toys, the initial decision boundary will no longer align with low-density regions. Subsequently, the perturbation is applied to unlabeled 2-D toys whose positions in feature space have shifted relative to the original. However, in theory, the categories of 2-D toys do not change after these perturbations. Based on the smoothness assumption, the model needs to make consistent predictions about the original 2-D toy and its perturbed version. Inconsistent predictions lead to penalties imposed by the consistency learning term.

As training progresses, the decision boundary is gradually refined and becomes less responsive to nonessential perturbation in the samples, under the cluster assumption. Specifically, the representation of the same category of 2-D toys in the feature space becomes more aggregated, the distance between different categories of 2-D toys increases, and the decision boundary becomes more distinct and more stable as a result.

B. Multiperspective Consistency Learning

Compared to natural images, RSLIs offer a broader observational perspective and contain more diverse and complex land cover information, which can introduce significant interference. Consequently, directly applying SSL methods designed for natural images to the LM often results in inadequate

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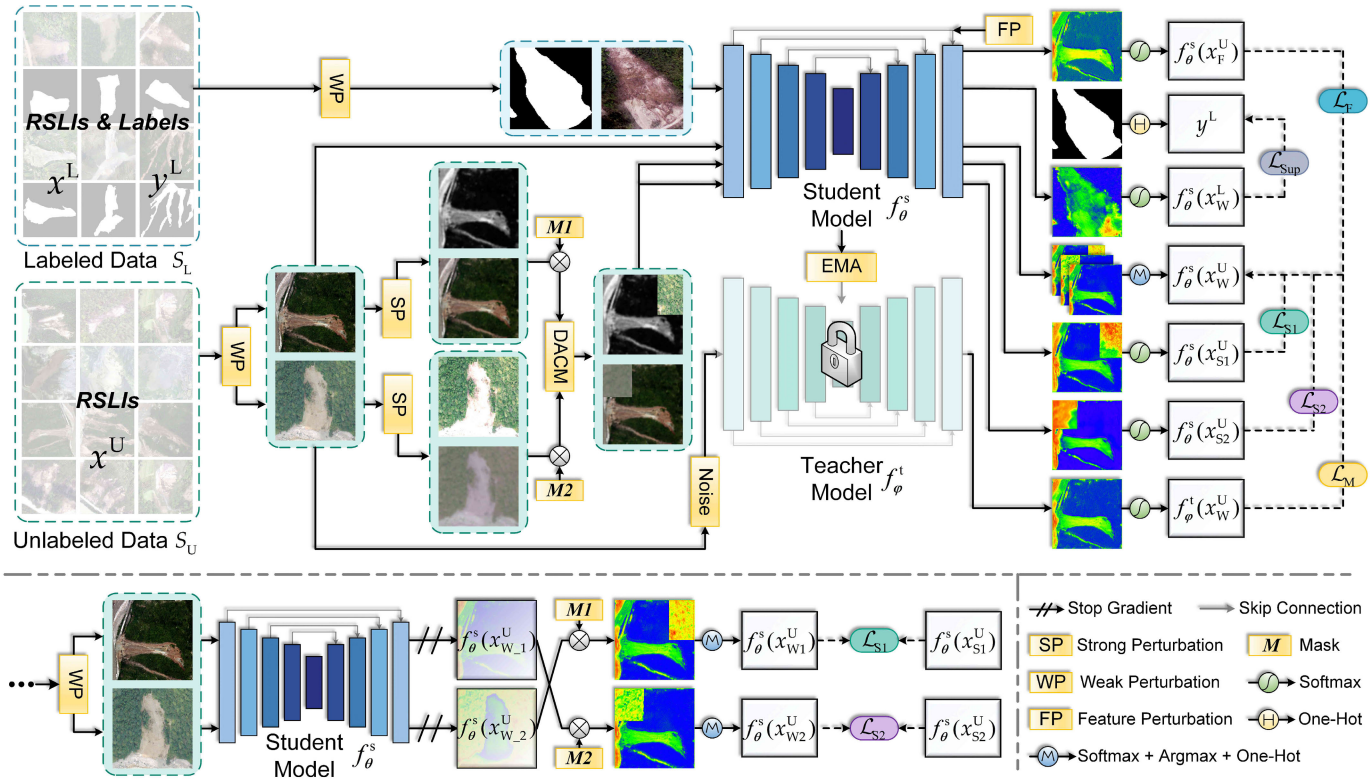


Fig. 6. Overview of the proposed Semi-LM. The framework is presented on the top half, with the detailed weak-to-strong consistency learning stream shown on the bottom half.

learning of the knowledge embedded in unlabeled data, leading to reduced accuracy or complete failure of SSL methods. Additionally, the quantity of RSLIs available for deep learning is significantly smaller than that of urban remote sensing images and natural images. This scarcity exacerbates the problem of model overfitting on the limited labeled data, generating pseudo-labels with low accuracy [44], which in turn degrades the effectiveness of consistency learning. Therefore, to expand and fully utilize the original perturbation space and to more effectively mine and learn knowledge from the numerous unlabeled RSLIs, it is essential to focus on the types and application methods of perturbations.

To address the aforementioned challenges, this article integrates MCL into a unified objective function by aligning the LM results of input, feature, and model perturbations with the weak perturbation pseudo-labels. This integration constructs the Semi-LM framework for LM and the overall structure is illustrated in Fig. 6. Semi-LM comprises two end-to-end landslide segmentation models: a student model f_θ^s and a teacher model f_φ^t. The student model learns knowledge through supervised learning (SL) and MCL stream. Specifically, in the SL stream, labeled RSLIs x^L undergo weak perturbation data augmentation and are then fed into the student model to obtain the LM results $f_\theta^s(x_W^L)$. Subsequently, the supervised loss $\mathcal{L}_{Sup}$ is computed as follows:

$$f_\theta^s\left(x_W^L\right) = f\left(\theta, \mathcal{A}^W\left(x^L\right)\right) \tag{2}$$

$$\mathcal{L}_{Sup} = \frac{1}{M_L}\sum_{i=1}^{M_L} \ell_{CE/Dice}\left(\text{One}-\text{Hot}\left(y_i^L\right), f_\theta^s\left(x_{W_i}^L\right)\right) \tag{3}$$

where $\mathcal{A}^W$ denotes weak perturbation data augmentation, which contains resizing, cropping, rotating by $n \times 90°$, flipping, and transposing. $f(\theta,\cdot)$ is a formulaic representation of student model and combine Softmax$(\cdot,\cdot)$ operation. $\ell_{CE/Dice}(\cdot,\cdot)$ represents the binary cross-entropy loss or Dice loss.

Unlabeled RSLIs are fed into the student model after applying weak perturbation to obtain hard pseudo-labels $f_\theta^s(x_W^U)$ Typically, models have higher quality LM results for weakly perturbed inputs than for strongly perturbed inputs, while the strong one can introduce additional information and break the original data distribution. Therefore, these hard pseudo-labels can serve as fitting targets for other unsupervised loss terms to achieve consistency learning. It is formulated as

$$f_\theta^s\left(x_W^U\right) = \text{One}-\text{Hot}\left(\text{Argmax}\left(f\left(\theta, \mathcal{A}^W\left(x^U\right)\right)\right)\right). \tag{4}$$

In the weak-to-strong consistency stream, unlabeled RSLIs are first applied to weak perturbation and strong perturbation data augmentation for input perturbation. During this process, the DACM is introduced to mitigate the disruption of both shallow and deep landslide features and microtopographic information. Then, the perturbed versions are fed into the student model to obtain $f_\theta^s(x_{S1}^U)$ and $f_\theta^s(x_{S2}^U)$, which are enforced to be alignment with the DACM-processed hard pseudo-labels $f_\theta^s(x_{W1}^U)$ and $f_\theta^s(x_{W2}^U)$ (Fig. 6, bottom). This process constructs the weak-to-strong consistency loss $\mathcal{L}_{S1}$ and $\mathcal{L}_{S2}$. The specific process can be described as follows:

$$f_\theta^s\left(x_{S1}^U\right) = f\left(\theta, \text{DACM}\left(\mathcal{A}^S\mathcal{A}^W\left(x_1^U\right), \mathcal{A}^S\mathcal{A}^W\left(x_2^U\right), M1\right)\right) \tag{5}$$

$$f_\theta^s\left(x_{W1}^U\right) = \text{DACM}\left(f_\theta^s\left(x_{W_1}^U\right), f_\theta^s\left(x_{W_2}^U\right), M1\right) \tag{6}$$

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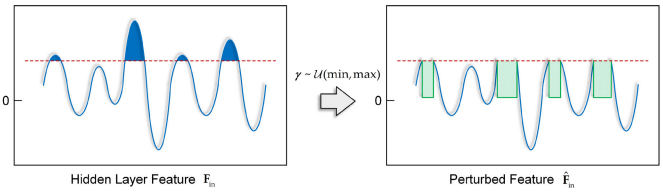


Fig. 7. Visualization of the F-Drop process.

$$\mathcal{L}_{S1} = \frac{1}{M_U} \sum_{i=1}^{M_U} \ell_{Dice}\left(f_\theta^s\left(x_{W1_i}^U\right), f_\theta^s\left(x_{S1_i}^U\right)\right) \tag{7}$$

where $\mathcal{A}^S$ denotes strong perturbation data augmentation, which consists of color jitter, blur, edge enhancement, and grayscale. $M1$ and $M2$ denote the binary mask. The calculation process of $\mathcal{L}_{S2}$ is consistent with $\mathcal{L}_{S1}$, only replacing the parameters $M1$ with $M2$.

The feature perturbation consistency learning stream involves applying perturbations to the hidden layer features $\mathbf{F}_{in}$ and enforcing the perturbed output $f_\theta^s(x_F^U)$ to be consistent with the hard pseudo-labels, thereby constructing the feature-perspective consistency loss $\mathcal{L}_F$. In this article, we adopt F-Drop [28] as the perturbation function, as illustrated in Fig. 7. We then apply it at the stage determined by the FPA strategy. The computation process for F-Drop and $\mathcal{L}_F$ can be represented as follows:

$$\hat{\mathbf{F}}_{in} = \mathbf{F}_{in} \odot \left\{ \left(\text{Norm}\left(\sum_{k=1}^{C} \mathbf{F}_{in_k} \right) \right) < \gamma \right\}_1 (\gamma \sim \mathcal{U}(\min, \max)) \tag{8}$$

$$\mathcal{L}_F = \frac{1}{M_U} \sum_{i=1}^{M_U} \ell_{Dice}\left(f_\theta^s\left(x_{W_i}^U\right), f_\theta^s\left(x_{F_i}^U\right)\right) \tag{9}$$

where $\hat{\mathbf{F}}_{in}$ represents the perturbed version of the feature map, which is fed into the subsequent model layers to obtain the feature perturbation LM result $f_\theta^s(x_F^U)$. $\odot$ is the pixel-by-pixel multiplication and Norm(,) represents the normalization operation. γ is a random threshold tensor of the same size as $\mathbf{F}_{in}$. The $\min, \max \in (0, 1)$ and $\{\text{condition}\}_1$ is a Boolean function that outputs 1 if the condition is true and 0 otherwise.

Different from feature perturbation that applies perturbations to hidden layer features, model perturbations target the structure and parameters of the model. The teacher–student model shares the same architecture but utilizes different weight initialization methods to introduce greater initialization diversity into the teacher–student framework. Additionally, the Gaussian noise is fed into the teacher model to simulate random noise or environmental disturbances that may occur in real-world scenarios. This noise operates synergistically with model perturbations to enhance the effectiveness of consistency regularization, compelling the student model to focus more on invariances and essential semantic information during feature extraction. Consequently, this approach improves the ability to leverage unlabeled RSLIs effectively in model. The teacher model outputs the soft pseudo-labels $f_\varphi^t(x_W^U)$, which guide the parameter updates of student model by constructing a model consistency loss $\mathcal{L}_M$. In turn, the student model improves the teacher model through EMA updates. The parameter update processes for $\mathcal{L}_M$ and teacher model are defined as follows:

$$f_\varphi^t\left(x_W^U\right) = f\left(\varphi, \text{Noise}\left(\mathcal{A}^W\left(x^U\right)\right)\right) \tag{10}$$

$$\mathcal{L}_M = \frac{1}{M_U} \sum_{i=1}^{M_U} \ell_{CE/Dice}\left(f_\theta^s\left(x_{W_i}^U\right), f_\varphi^t\left(x_{W_i}^U\right)\right) \tag{11}$$

$$\varphi_T = \alpha\varphi_{T-1} + (1-\alpha)\theta_T \tag{12}$$

where Noise(,) $\in (-0.1, 0.1)$ denotes the Gaussian noise and $\alpha = 0.999$ is the smoothing coefficient hyperparameter. θ_T is the student model parameters at T moment, and φ_T and φ_{T-1} represent the teacher model parameters at T and $T-1$ moments, respectively.

To ensure that the model can learn beneficial knowledge for LM from each learning stream, all learning streams are independently implemented. Therefore, the overall loss function $\mathcal{L}$ can be expressed as follows:

$$\mathcal{L} = \frac{1}{M_U} \sum_{i=1}^{M_U} \frac{\left(\mathcal{L}_{Sup}^i + \beta\mathcal{L}_{S1}^i + \delta\mathcal{L}_{S2}^i + \eta\mathcal{L}_F^i + \mu\mathcal{L}_M^i\right)}{2}. \tag{13}$$

To simplify the model and avoid parameter bias, we chose to apply equal weights to all unsupervised learning streams, i.e., $\beta = \delta = \eta = \mu = 0.25$. To provide a clearer description and understanding of the Semi-LM workflow, its training process is listed in Algorithm 1.

Algorithm 1 Semi-LM in a PyTorch-Like Style

Input: labeled RSLIs dataset $S_L = \{(x_i^L, y_i^L)\}_{i=1}^{M_L}$; unlabeled RSLIs dataset $S_U = \{(x_i^U)\}_{i=1}^{M_U}$

student model f_θ^s; teacher model f_φ^t //same structure, different initialization
weak perturbation $\mathcal{A}^w$; strong perturbation $\mathcal{A}^s$
for T in [1, $Epoch$] **do**
  for each minibatch B do
    $f_\theta^s(x_W^L) = f(\theta, \mathcal{A}^w(x^L))$
    $\mathcal{L}_{Sup} = \ell_{ce/Dice}(\text{One} - \text{Hot}(y^L), f_\theta^s(x_w^L))$ //supervised learning stream
    $f_\theta^s(x_w^U) = \text{One} - \text{Hot}(\text{Argmax}(f(\theta, \mathcal{A}^w(x^U))))$
    $f_\theta^s(x_{s1}^u) = f(\theta, \text{DACM}(\mathcal{A}^s\mathcal{A}^w(x_1^u), \mathcal{A}^s\mathcal{A}^w(x_2^u), M1))$
    $f_\theta^s(x_{w_1}^U) = \text{DACM}(f_\theta^s(x_{W_1}^U), f_\theta^s(x_{W_2}^U), M1)$
    $f_\theta^s(x_{s2}^U) = f(\theta, \text{DACM}(\mathcal{A}^s\mathcal{A}^w(x_1^U), \mathcal{A}^s\mathcal{A}^w(x_2^U), M2))$
    $f_\theta^s(x_{W_2}^U) = \text{DACM}(f_\theta^s(x_{W_1}^U), f_\theta^s(x_{W_2}^U), M2)$
    $\mathcal{L}_{Sl} = \ell_{Dice}(f_\theta^s(x_{W1}^U), f_\theta^s(x_{S1}^U))$ //weak-to-strong consistency learning stream 1
    $\mathcal{L}_{S2} = \ell_{Dice}(f_\theta^s(x_{w2}^U), f_\theta^s(x_{s2}^U))$ //weak-to-strong consistency learning stream 2
    $f_\theta^s(x_F^U) = f(\boldsymbol{\theta}, \mathcal{A}^W(x^U), \mathbf{M}_{\text{F-Drop}})$
    $\mathcal{L}_F = \ell_{Dice}(f_\theta^s(x_W^U), f_\theta^S(x_F^U))$ //feature consistency learning stream
    $f_\varphi^t(x_w^U) = f(\varphi, \text{Noise}(\mathcal{A}^W(x^U)))$
    $\mathcal{L}_M = \ell_{CE/Dice}(f_\theta^s(x_w^U), f_\varphi^t(x_W^U))$ //model consistency learning stream
    $\mathcal{L} = (\mathcal{L}_{Sup} + 0.25 \times \mathcal{L}_{S1} + 0.25 \times \mathcal{L}_{S2} + 0.25 \times \mathcal{L}_F + 0.25 \times \mathcal{L}_M)/2$ // overall loss
    Optimizer($\mathcal{L}$) //update θ using optimizer
    $\varphi_T = \alpha\varphi_{T-1} + (1-\alpha)\theta_T$ //update φ through EMA
  end for
end for
return f_θ^s, f_φ^t

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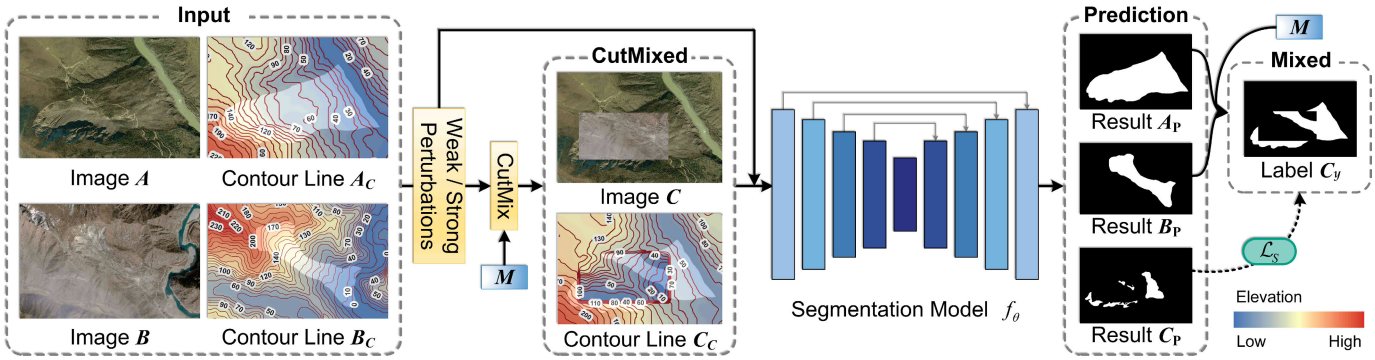


Fig. 8. Flowchart illustrating the application of the CutMix to the LM model training process. Contour lines are generated from the DEM and are not included as mandatory inputs.

C. Simple and Effective DACM

The weak-to-strong consistency regularization-based SSL has achieved booming success in semantic segmentation [18], [25]. Its success lies in the fact that weak perturbation can generate accurate predictions, while strong perturbation introduces additional information and mitigates confirmation bias.

To provide strong and varied perturbations in a strongly augmented view, CutMix [27] and its variants [68], [69] have been introduced. CutMix aims to generate a hybrid training sample C and corresponding mixed label C_y by combining original training samples A and B. This process can be formulated as follows:

$$C_{\mathrm{P}} = f\left(\theta, \mathbf{M} \odot \mathcal{A}^{\mathrm{S}}\mathcal{A}^{\mathrm{W}}(A) + (\mathbf{1}-\mathbf{M}) \odot \mathcal{A}^{\mathrm{S}}\mathcal{A}^{\mathrm{W}}(B)\right) \quad (14)$$

$$C_y = \mathbf{M} \odot f\left(\theta, \mathcal{A}^{\mathrm{W}}(A)\right) + (\mathbf{1}-\mathbf{M}) \odot f\left(\theta, \mathcal{A}^{\mathrm{W}}(B)\right) \quad (15)$$

$$\mathcal{L}_{\mathrm{S}} = \frac{1}{M_{\mathrm{U}}} \sum_{i=1}^{M_{\mathrm{U}}} \ell_{\mathrm{Dice}}\left(C_y, C_{\mathrm{P}_i}\right) \quad (16)$$

where C_{P} represents the LM result. $\mathbf{M} \in \{0,1\}^{H \times W}$ is a binary mask that defines the location of deletion and retention of the two images, the deleted area overrides 0, and the retained area overrides 1. This 1 represents a mask filled with ones that are the same size as $\mathbf{M}$.

The randomly generated binary mask in CutMix can introduce a diverse and rich variety of input perturbations, but it can also disrupt both shallow and deep features and micro-topographic information of RSLIs. In Fig. 8, we illustrate the application of CutMix in the training process of the LM model. The input data include an optical image and optionally digital elevation model (DEM). The contour lines in Fig. 8 are generated from the DEM for better understanding and observation, but the actual input should remain as DEM. Obtaining the landslide microtopographic features from the image A, such as the main scarp, minor scarp, toe, and cracks. Then, by combining it with the characteristics in the contour line A_C, where the landslide leading edge protrudes upward and the deposition area at the rear slopes downward, we can accurately delineate the landslide boundary. In the image B, the optical image alone provides limited information, but with the aid of DEM data, we can improve the LM accuracy. However, when CutMix is applied, we can observe that the original landslide features are significantly disrupted in the optical image C. In the new contour line C_C, there are dense overlapping lines, which geographically suggest a cliff, contradicting typical landslide characteristics. The weak-to-strong consistency learning forces C_{P} and C_y to remain consistent. This leads to a contradiction in knowledge learning compared to those obtained through other perturbation consistency learning streams. Such contradiction may cause the model to learn incorrect information, ultimately reducing the accuracy of LM.

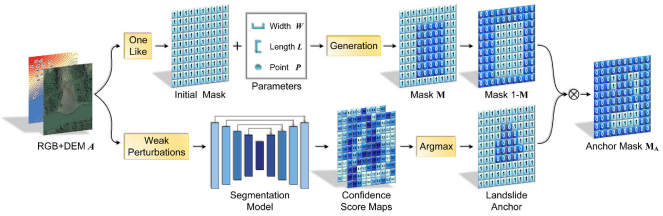


Fig. 9. Illustration of the anchor mask generation process.

To address the limitation of CutMix in preserving the semantic information of landslides, we propose enhancing the binary mask generation process with an anchor positioning branch, resulting in the DACM. The goal of DACM is to mitigate the risk of overfitting specific patterns in labeled data. It enforces consistency between the two strong perturbation views and a weak perturbation view, allowing DACM to leverage the advantages of contrastive learning, thereby facilitating the learning of more discriminative feature representations from unlabeled RSLIs. In addition, the DACM will generate the NO-Cut zone mask for the CutMix, minimizing any potential harm to the landslide. The process of generating the anchor mask is illustrated in Fig. 9, and it consists of two branches.

In the mask generation branch, we get an initial mask of the same size as the input RSLI and filled with ones. Then, the mask $\mathbf{M}$ is generated based on hyperparameters. In the anchor positioning branch, the segmentation model takes in A as input after weak perturbation to get the confidence score maps. This map is then processed through the Argmax operation to obtain a landslide anchor, where zero represents the landslide and one represents the background. Finally, the anchor mask $\mathbf{M_A}$ is obtained by the Hadamard product of $\mathbf{1}-\mathbf{M}$ and landslide

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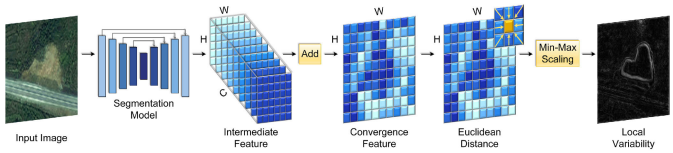


Fig. 10. Flowchart of measuring local variability process.

anchor. In the training process, the confidence score map is obtained from the original image prediction process in CutMix, so the generation of $\mathbf{M_A}$ will not add extra computational burden. It is formulated as

$$\mathbf{M_A} = 1 - \mathbf{M} \odot \mathrm{Argmax}\left(f\left(\mathcal{A}^{\mathrm{W}}(A)\right)\right). \tag{17}$$

D. FPA Strategy of LM

Inspired by that feature maps can capture more discriminative contextual information and introduce feature perturbations for consistency training [29], we integrate feature perturbations into LM. The effectiveness of consistency training is often attributed to the clustering assumption, which posits that decision boundaries should lie in low-density regions to prevent the misclassification of data from the same class into different categories. Specifically, the minimization of the consistency learning stream encourages the decision boundary for the unsupervised sample regions to become more flattened, thereby moving decision boundaries closer to low-density regions. This approach helps to maintain consistent predictions and improves the robustness of the model in handling unlabeled data.

Thus, identifying the stage with the highest overlap between low-density regions and class boundaries ensures the effectiveness of feature perturbation consistency learning. Previous studies have suggested applying feature perturbation at the encoder output [28], [70] or the encoder intermediate stage [29] to achieve better perturbation effects. However, the significant differences between RSLIs and natural images or urban remote sensing images, and the suggestions limit the application effectiveness in LM tasks. To address this, we analyze the clustering assumption through local variability in the LM process and propose the FPA strategy tailored for LM.

Local variability is computed by averaging the Euclidean distances between a central patch and its eight surrounding patches, as illustrated in Fig. 10. The category index variation of patches inside the ground object is 0, corresponding to low local variability. In contrast, the category index of patches at ground object junctions varies significantly and shows noticeable variations of local variability, describing the presence of low-density regions [28].

Fig. 11 illustrates the local variability measurement results in the LM process using U-Net. It can be observed that during stages No.2, No.3, and No.4, the overlap between class boundaries and low-density regions is relatively low, which does not align with the clustering assumption and ultimately reduces or negates the effectiveness of consistency regularization. In contrast, the decoding stage, by integrating shallow features from the encoding stage and progressively recovering

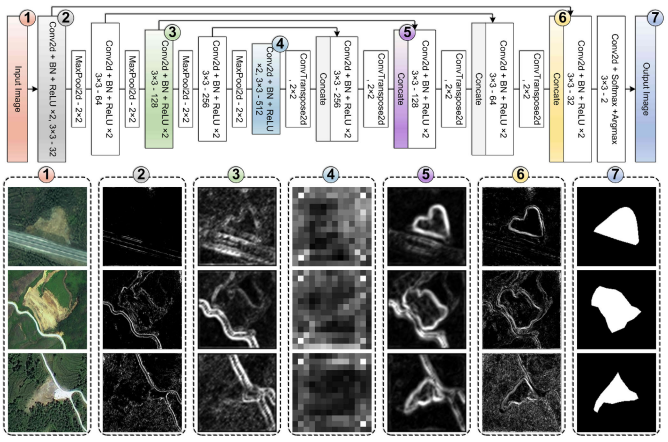


Fig. 11. Cluster assumption in consistency regularization-based SSL LM. Examples from No.1 RSLIs, No.2 and No.3 intermediate layer in the encoder, No.4 the output of the encoder, No.5 and No.6 intermediate layer in the decoder, and No.7 LM results.

spatial resolution and detail features, retains higher order semantic information while recovering more detailed features. Therefore, the class boundaries in the decoder stage exhibit a higher overlap with low-density regions. Additionally, the No.5 local variability result includes some low-density regions at the boundaries of nonlandslide objects. These low-density regions may interfere with consistency regularization, increasing the risk of misclassification. This observation motivates the FPA strategy where feature perturbations are assigned to stage No.6, which are the outputs of the decoder in the LM task.

IV. Experiments and Analysis

In this section, we introduce and compare the five open-source landslide datasets used in the experiments in Section IV-A and present the details of the experimental implementation and evaluation metrics in Section IV-B. To verify the effectiveness of the proposed Semi-LM framework, we select SL and six SOTA SSL methods—MT [63], CCT [28], CPS [71], U^{2}PL [62], UniMatch [18], WSCL [25], and SegMind [58] for comparative experimental analysis in Section IV-C. It is important to note that WSCL and SegMind are specifically designed for remote sensing image segmentation, while the other models are used for natural image segmentation. Subsequently, in Section IV-D, we conduct ablation experiments and report that the DACM method and FPA strategy, along with MCL, can effectively improve the LM accuracy. Finally, in Section IV-E, we study the computational load of Semi-LM and its performance under different base models.

A. Dataset

To assess the efficacy of this framework, we conduct comprehensive experiments on five widely used datasets, including Sichuan [72], Bijie [73], Nepal [4], Luding [74], and LRSTTC [8]. These five datasets consist of remote sensing images from different regions around the world and include both coseismic and rainfall-induced landslides. Detailed information about these datasets is provided in Table I and Fig. 12.

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TABLE I
DETAILED INFORMATION OF THREE LANDSLIDE DATASETS

Dataset	Data type	Data source	Spatial resolution	Training	Validation	Test
Sichuan	Optical image	Aerial imagery	0.2-0.9m	70	15	22
Bijie	Optical image and DEM	TripleSat	0.8m	514	44	212
Nepal	Optical image	Landsat 8	30m	200	35	40
Luding	Optical image	GF-6	2m	140	20	40
LRSTTC	Optical image	Google Earth	2m	444	74	222

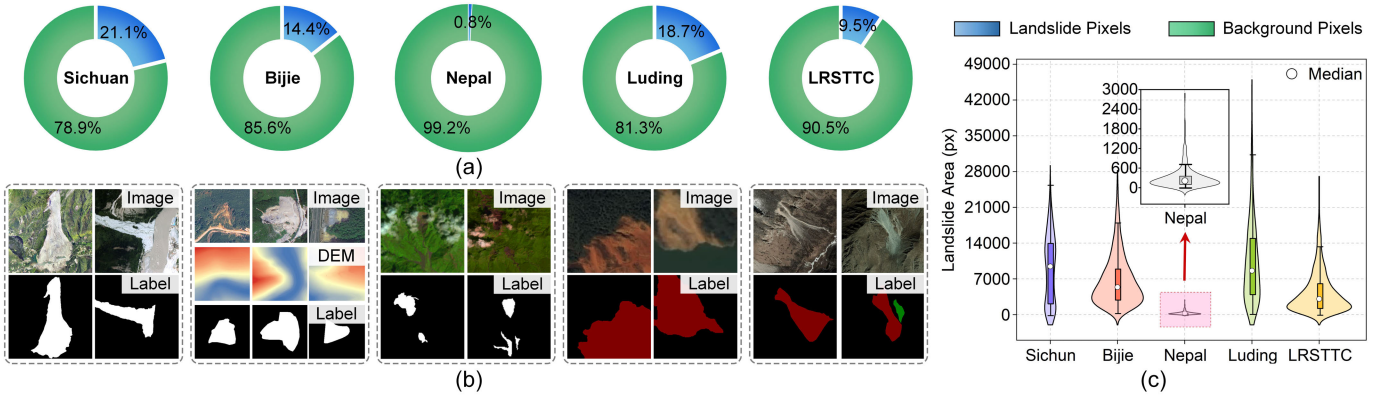


Fig. 12. Statistical information of the experimental datasets and presentation of some samples. (a) Ratio of landslide pixels and background pixels of datasets. (b) Dataset samples. (c) Landslide size and distribution statistics.

1) The Sichuan dataset includes earthquake and rainfall-induced landslides, and debris flow disasters. It comprises 59 landslide images and 48 debris flow images. The spatial coverage includes the areas affected by the "5 · 12" Wenchuan earthquake, "4 · 20" Lushan earthquake, and "8 · 8" Jiuzhaigou earthquake, as well as regions along the Jinsha River and Dadu River.
2) The Bijie dataset covers Bijie in Southwest China with a focus on rainfall-induced landslides. In addition to optical images, it provides DEM data with a spatial resolution of 2 m to assist in LM.
3) The Nepal dataset includes landslides that are distributed across Nepal, primarily triggered by the two large earthquakes that struck Gorkha on April 25 and Dolakha on May 12, 2015. The false-color images are composed of bands 6-5-4 from Landsat 8.
4) The Luding dataset is composed of landslides triggered by the earthquake that occurred on September 5, 2022, in Luding, Sichuan, China. All data have been artificially resampled to a spatial resolution of 2 m.
5) The LRSTTC dataset includes landslides along the Sichuan–Tibet Railway from Sichuan to Tibet, triggered by various factors. Notably, the dataset is divided into old and new landslides. In this work, we have selected the old landslides for experimentation.

Due to differences in sensors and geographical locations, the characteristics of the five datasets exhibit variations. As illustrated in Fig. 12(a), the Sichuan dataset has the highest proportion of landslide pixels relative to total ground pixels. Conversely, the Nepal dataset has the lowest proportion of landslide pixels, at only 0.8%, suggesting that the landslide scales in this dataset are smaller and more challenging to detect. Additionally, as shown in Fig. 12(c), the Luding dataset exhibits the most pronounced variation in landslide scales, indicating significant multiscale effects within this dataset. In contrast, the landslide size variability is minimal in the Nepal dataset. These distinct characteristics across datasets validate the generalizability and effectiveness of Semi-LM.

B. Experimental Setup and Evaluation Metrics

All experiments are implemented in an environment of Python3.8 + Pytorch1.12.1 + CUDA11.3, and an NVIDIA GeForce RTX 3060 12-GB memory GPU is used to speed up the model training. All data are resampled to 256 × 256. The input batch size of SSL is 16, i.e., eight labeled images and eight unlabeled images. The SL batch size is 8. No pretraining weights are used for model training. The basic model for comparison and ablation experiments is U-Net, and ResNet50-based DeepLabV3Plus is validated during the discussion. The training epoch of the experimental groups is determined based on the number of labeled images, and all the training processes converged. Each experiment is repeated three times, and the mean and standard deviation are taken as the final results.

Five evaluation metrics are selected to evaluate the experimental results, including overall accuracy (OA), precision (P), recall (R), F_1-Score, and intersection over union (IoU). The evaluation metrics are calculated from TP, FP, FN, and TN, as described in Table II.

C. Comparison With Existing Alternatives

1) Results on the Sichuan Dataset: As shown in Table III, Semi-LM outperforms OA, IoU, and F_1-Score across the three partition protocols. Notably, in the 14% partition protocol, the IoU and F_1-Score of Semi-LM are 13.2% and 11.6%

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TABLE II
CONFUSION MATRIX FOR LM RESULTS

		Actual Classification	
		Landslide	Background
Predicted Classification	Landslide	TP	FP
	Background	FN	TN

TABLE III
COMPARISONS BETWEEN SEMI-LM WITH SOTA METHODS ON THE SICHUAN DATASET

Partition Protocol	Methods	OA /%	Precision /%	Recall /%	IoU /%	F_1-Score /%
7% (5)	SL	79.02±0.0	51.34±0.1	63.85±0.0	39.78±0.0	56.92±0.0
	MT	79.71±0.7	52.48±1.2	71.15±2.1	43.23±0.2	60.37±0.2
	CCT	83.16±0.8	60.57±2.4	65.20±5.6	45.61±2.4	62.61±2.2
	CPS	86.78±0.1	**74.06±0.2**	60.17±0.5	49.69±0.4	66.39±0.3
	U^2PL	81.22±0.5	55.15±1.1	72.76±1.6	45.68±0.1	62.71±0.1
	UniMatch	82.70±1.0	59.15±2.2	65.81±2.1	45.27±2.2	62.29±2.1
	SgeMind	82.38±0.1	58.10±0.2	67.50±0.5	45.40±0.4	62.44±0.3
	WSCL	80.27±0.2	53.11±0.4	**77.84±0.5**	46.13±0.3	63.14±0.3
	Semi-LM	**87.21±0.2**	70.31±1.4	71.25±1.6	**54.74±0.2**	**70.75±0.2**
14% (10)	SL	86.39±0.7	78.77±0.6	51.11±4.3	44.88±3.3	61.89±3.1
	MT	87.48±0.2	79.33±0.5	57.22±1.0	49.80±0.9	66.48±0.8
	CCT	88.26±0.3	76.43±1.4	**66.50±2.3**	55.15±1.5	71.08±1.2
	CPS	89.34±0.0	**90.56±0.0**	56.83±0.1	53.65±0.1	69.84±0.1
	U^2PL	87.11±0.3	74.96±1.5	61.03±0.3	50.69±0.5	67.27±0.5
	UniMatch	88.56±0.7	80.37±3.6	62.87±3.5	54.40±2.4	70.43±2.0
	SgeMind	88.38±0.2	84.93±2.5	56.64±1.0	51.43±0.1	67.92±0.1
	WSCL	89.17±0.0	86.54±1.3	59.37±1.3	54.32±0.5	70.40±0.4
	Semi-LM	**89.95±0.0**	86.01±1.0	64.16±0.7	**58.08±0.1**	**73.49±0.1**
28% (20)	SL	88.43±0.6	77.49±3.3	66.49±6.3	55.42±3.1	71.26±2.6
	MT	89.88±0.2	79.41±0.5	72.07±0.5	60.72±0.6	75.56±0.5
	CCT	90.26±0.5	80.04±2.2	73.51±0.1	62.12±1.4	76.62±1.0
	CPS	89.92±0.0	79.86±0.6	71.63±0.4	60.67±0.0	75.52±0.0
	U^2PL	89.49±0.2	78.19±1.0	71.58±0.2	59.67±0.5	74.74±0.4
	UniMatch	88.98±0.4	73.07±2.2	78.33±2.8	60.69±0.8	75.53±0.6
	SgeMind	90.22±0.5	79.19±1.5	74.58±1.0	62.36±1.5	76.80±1.2
	WSCL	90.21±0.6	80.59±3.5	72.73±3.5	61.72±1.8	76.31±1.4
	Semi-LM	**91.34±0.1**	**81.03±0.9**	**78.52±0.4**	**66.32±0.4**	**79.75±0.3**
100%	SL	90.37±0.4	86.05±1.9	66.48±1.6	60.00±1.5	74.98±1.2

*** Supervised Learning (SL), Mean Teacher (MT).**

TABLE IV
COMPARISONS BETWEEN SEMI-LM WITH SOTA METHODS ON THE BIJIE DATASET

Partition Protocol	Methods	OA /%	Precision /%	Recall /%	IoU /%	F_1-Score /%
2% (10)	SL	88.19±0.2	66.32±1.3	40.83±0.8	33.82±0.8	50.54±0.9
	MT	90.27±0.0	73.94±0.4	52.79±0.4	44.50±0.1	61.60±0.1
	CCT	89.31±0.8	65.06±4.8	61.38±2.2	45.97±1.2	62.98±1.1
	CPS	91.66±0.1	**90.49±0.2**	48.62±0.7	46.26±0.7	63.25±0.7
	U^2PL	90.82±0.1	79.94±1.4	50.56±0.1	44.86±0.3	61.94±0.3
	UniMatch	92.62±0.0	81.07±1.5	65.38±1.9	56.67±0.7	72.34±0.6
	SgeMind	90.34±0.8	66.73±4.3	70.61±3.5	51.97±0.8	68.39±0.7
	WSCL	92.03±0.2	83.17±0.8	57.80±2.1	51.73±1.6	68.17±1.3
	Semi-LM	**93.28±0.0**	79.71±0.3	**73.13±0.2**	**61.65±0.2**	**76.28±0.2**
6% (30)	SL	92.25±0.3	80.20±3.0	63.38±2.4	54.73±1.5	70.73±1.3
	MT	92.66±0.0	77.37±0.8	71.16±0.7	58.89±0.1	74.13±0.1
	CCT	92.73±0.2	78.20±0.4	70.44±1.4	58.88±1.2	74.11±1.0
	CPS	93.11±0.1	**84.93±0.0**	64.89±0.8	58.19±0.7	73.57±0.5
	U^2PL	92.80±0.1	81.65±0.7	66.12±1.0	57.56±0.8	73.06±0.6
	UniMatch	93.25±0.5	84.80±2.9	66.27±0.8	59.25±2.0	74.39±1.6
	SgeMind	91.66±0.5	69.79±3.3	77.60±3.2	57.92±1.0	73.35±0.8
	WSCL	92.85±0.1	77.49±0.9	72.78±0.3	60.08±0.4	75.06±0.3
	Semi-LM	**93.49±0.0**	79.25±1.0	**75.79±0.9**	**63.23±0.0**	**77.47±0.0**
20% (104)	SL	93.52±0.1	81.12±4.7	74.06±5.2	62.75±1.2	77.10±0.9
	MT	94.06±0.0	86.11±0.1	71.33±0.2	63.97±0.1	78.03±0.1
	CCT	94.18±0.3	83.14±1.4	76.08±2.7	65.87±1.8	79.41±1.3
	CPS	94.45±0.0	85.50±0.1	75.16±0.1	66.67±0.2	80.00±0.1
	U^2PL	94.12±0.0	86.00±1.2	71.96±0.8	64.40±0.1	78.34±0.1
	UniMatch	94.58±0.1	**89.90±1.6**	71.42±2.0	66.07±1.0	79.56±0.7
	SgeMind	94.48±0.1	86.64±1.1	74.08±1.6	66.45±0.7	79.85±0.5
	WSCL	94.44±0.0	86.08±1.2	74.48±1.9	66.44±0.7	79.83±0.5
	Semi-LM	**94.96±0.0**	85.20±0.4	**79.79±0.1**	**70.08±0.2**	**82.41±0.1**
30% (154)	SL	94.52±0.3	86.44±1.6	74.64±2.2	66.81±2.0	80.08±1.4
	MT	94.91±0.0	88.49±0.3	75.35±0.3	68.62±0.3	81.39±0.2
	CCT	95.08±0.1	88.62±1.1	76.62±2.3	69.71±1.3	82.14±0.9
	CPS	94.87±0.0	86.09±0.9	77.93±1.8	69.18±0.8	81.78±0.6
	U^2PL	95.13±0.0	88.64±1.0	76.96±1.1	70.03±0.2	82.37±0.2
	UniMatch	95.12±0.0	**90.49±0.7**	74.83±1.2	69.36±0.6	81.91±0.4
	SgeMind	95.19±0.1	86.84±1.0	79.50±1.8	70.92±0.8	82.98±0.5
	WSCL	94.78±0.4	86.25±4.3	77.27±1.5	68.67±1.5	81.42±1.1
	Semi-LM	**95.82±0.0**	88.08±0.3	**82.94±0.4**	**74.57±0.1**	**85.43±0.1**
100%	SL	95.65±0.0	87.38±0.8	82.46±1.2	73.67±0.4	84.83±0.3

*** Supervised Learning (SL), Mean Teacher (MT).**

higher than the SL baseline, respectively. Compared to the second-best SSL method, CCT, the F_1-Score of Semi-LM improves from 71.08% to 73.49%. Although Semi-LM does not achieve the highest precision and recall in the 7% and 14% partition protocols, it provides a better balance between the two. Specifically, CPS achieves the highest precision but at the cost of the lowest recall in the 7% partition protocol and the third-lowest recall in the 14% partition protocol. On the other hand, in the 14% partition protocol, the recall of CCT is 2.34% higher than Semi-LM, but the precision is 9.58% lower than Semi-LM. Fig. 13 shows some visualization results. It can be observed that Semi-LM more accurately identifies landslides and debris flows. This observation underscores the generalization capability of the proposed Semi-LM.

2) Results on the Bijie Dataset: Table IV compares the Semi-LM with other advanced methods on the Bijie dataset. Under four different partition protocols, Semi-LM achieves the highest scores in all metrics except Precision. Notably, under the 2% partition protocol, IoU and F_1-Score of Semi-LM are 27.83% and 25.74% higher than the SL baseline, respectively. Compared to the UniMatch, Semi-LM improves the IoU and F_1-Score by 4.98% and 3.94%, respectively. Additionally, it is noteworthy that under the 30% partition protocol, the performance of Semi-LM is comparable to that of SL under the 100% labeled partition protocol. This demonstrates the potential of Semi-LM in leveraging unlabeled data. Fig. 14 shows some of the LM visual results. As can be seen, the results of Semi-LM are closer to the actual ground conditions.

3) Results on the Nepal Dataset: Table V presents the comparative results on the Nepal dataset. Under the 15% partition protocol, Semi-LM achieves the maximum improvement of 7.41% in IoU and 8.64% in F_1-Score over the SL baseline. Although CPS attains the highest precision of 64.09% under the 40% partition protocol, its recall of 46.4% is the lowest among all compared methods. This indicates that a significant number of landslide instances are missed, which is unacceptable in practical land hazard surveys. In contrast, Semi-LM, while sacrificing some accuracy, detects more landslides, which helps reduce potential losses caused by geological

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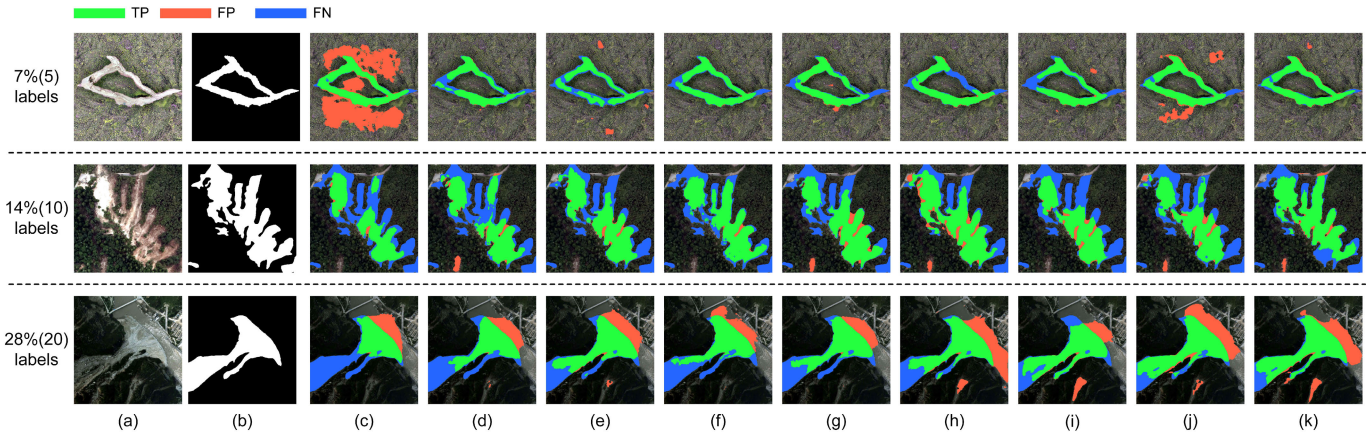


Fig. 13. LM visual comparison of Semi-LM with SOTA methods on the Sichuan dataset. (a) Input RSLIs. (b) Label. (c) SL. (d) MT. (e) CCT. (f) CPS. (g) U^{2}PL. (h) UniMatch. (i) SegMind. (j) WSCL. (k) Semi-LM.

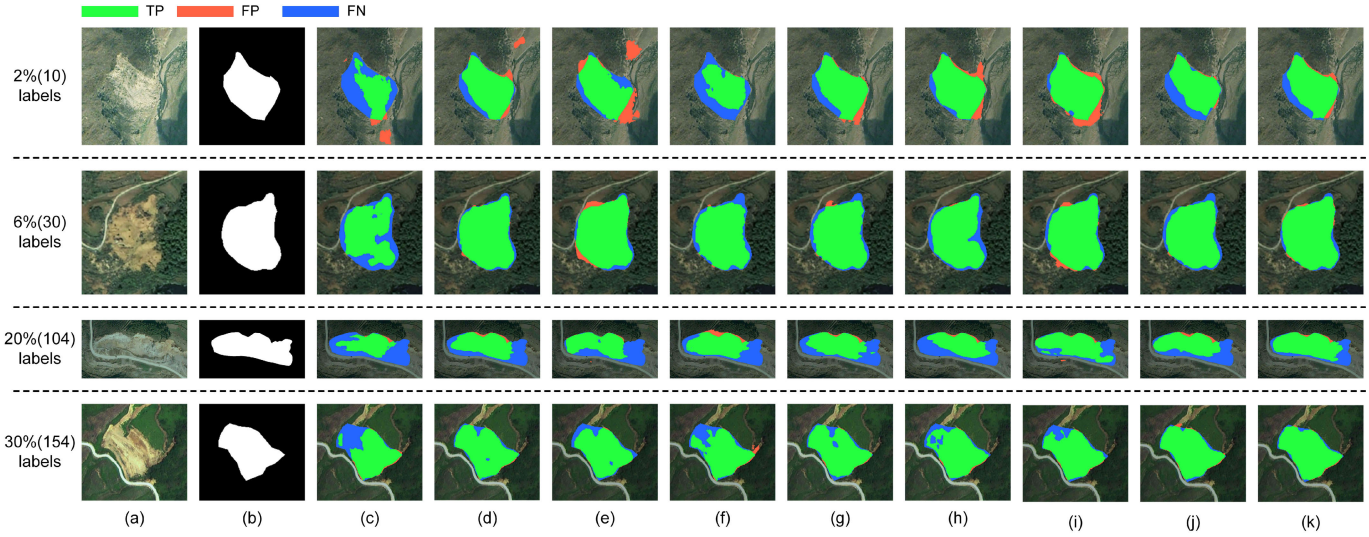


Fig. 14. LM visual comparison of Semi-LM with SOTA methods on the Bijie dataset. (a) Input RSLIs. (b) Label. (c) SL. (d) MT. (e) CCT. (f) CPS. (g) U^{2}PL. (h) UniMatch. (i) SegMind. (j) WSCL. (k) Semi-LM.

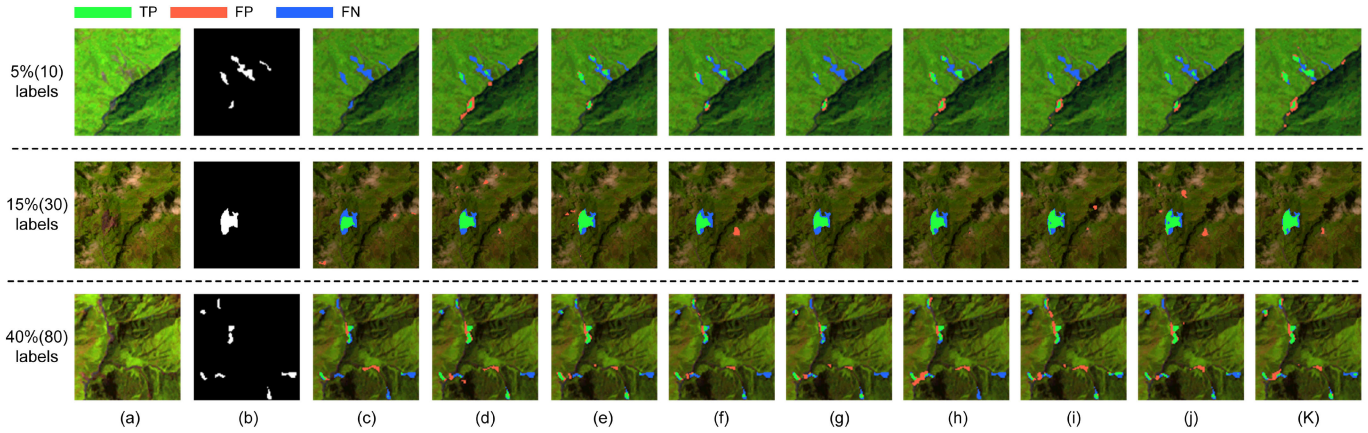


Fig. 15. LM visual comparison of Semi-LM with SOTA methods on the Nepal dataset. (a) Input RSLIs. (b) Label. (c) SL. (d) MT. I CCT. (f) CPS. (g) U^{2}PL. (h) UniMatch. (i) SegMind. (j) WSCL. (k) Semi-LM.

hazards. Fig. 15 shows the partial visualization results of LM. As illustrated, the landslides in the Nepal dataset are relatively small in scale. Due to the low image resolution, it is challenging to visually distinguish them from ground features. Nonetheless, Semi-LM still demonstrates superior recognition performance.

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TABLE V
COMPARISONS BETWEEN SEMI-LM WITH SOTA METHODS ON THE NEPAL DATASET

Partition Protocol	Methods	OA /%	Precision /%	Recall /%	IoU /%	F_1-Score /%
5% (10)	SL	99.15±0.0	48.77±1.4	30.77±1.5	23.26±1.2	37.73±1.6
	MT	98.89±0.0	36.62±1.3	42.86±0.7	24.59±0.4	39.47±0.5
	CCT	99.10±0.0	46.42±5.6	36.71±1.9	25.58±1.3	40.72±1.6
	CPS	99.11±0.0	**46.61±4.9**	35.56±1.2	25.21±1.8	40.24±2.3
	U^{2}PL	99.04±0.0	42.63±3.6	36.77±1.5	24.50±0.9	39.34±1.2
	UniMatch	99.13±0.0	47.25±1.1	32.24±2.0	23.66±0.8	38.26±1.0
	SgeMind	99.09±0.1	44.96±2.1	38.04±3.3	25.92±1.8	41.13±2.2
	WSCL	99.15±0.0	49.67±2.7	34.29±1.7	25.38±0.8	40.48±1.0
	Semi-LM	**98.99±0.0**	41.75±0.4	**50.34±1.7**	**29.56±0.6**	**45.63±0.7**
15% (30)	SL	99.12±0.0	48.10±5.5	39.75±2.7	27.62±1.6	43.25±2.0
	MT	99.06±0.0	44.27±2.3	46.26±1.3	29.18±0.7	45.18±0.8
	CCT	99.21±0.0	54.71±4.5	41.10±4.1	30.34±0.6	46.55±0.7
	CPS	99.20±0.0	53.09±1.6	41.17±3.6	30.07±1.3	46.22±1.5
	U^{2}PL	**99.25±0.0**	**58.83±1.4**	37.13±1.6	29.43±0.7	45.47±0.8
	UniMatch	99.19±0.0	52.66±5.1	41.34±3.1	30.01±2.0	46.12±2.4
	SgeMind	99.14±0.1	48.94±2.0	43.98±3.6	30.06±1.5	46.21±1.8
	WSCL	99.22±0.1	57.92±9.5	43.13±7.3	31.66±1.3	48.08±1.5
	Semi-LM	99.23±0.0	54.58±0.4	**49.45±0.4**	**35.03±0.2**	**51.89±0.2**
40% (80)	SL	99.24±0.0	55.36±1.9	47.07±2.4	34.12±1.7	50.85±1.9
	MT	99.18±0.0	51.23±1.5	53.62±0.6	35.48±0.5	52.37±0.5
	CCT	99.28±0.0	58.05±2.3	50.68±1.7	37.05±1.0	54.06±1.0
	CPS	**99.33±0.0**	**64.09±2.3**	46.40±1.2	36.82±1.2	53.81±1.3
	U^{2}PL	99.30±0.0	60.90±3.1	48.03±1.6	36.62±0.4	53.61±0.4
	UniMatch	99.24±0.0	55.61±2.9	50.10±2.1	35.81±2.1	52.70±2.3
	SgeMind	99.26±0.0	56.80±2.9	50.39±0.7	36.38±0.8	53.35±0.9
	WSCL	99.31±0.0	61.31±0.8	47.53±2.3	36.52±1.1	53.49±1.2
	Semi-LM	99.23±0.0	53.77±0.6	**61.04±1.2**	**40.02±0.4**	**57.16±0.4**
100%	SL	99.25±0.0	55.58±3.8	58.72±3.7	39.90±2.8	56.99±2.9

*** Supervised Learning (SL), Mean Teacher (MT).**

TABLE VI
COMPARISONS BETWEEN SEMI-LM WITH SOTA METHODS ON THE LUDING DATASET

Partition Protocol	Methods	OA /%	Precision /%	Recall /%	IoU /%	F_1-Score /%
3% (5)	SL	94.19±0.0	78.28±0.2	**88.59±0.1**	71.11±0.1	83.12±0.0
	MT	95.02±0.1	82.53±0.9	87.72±0.4	73.97±0.4	85.04±0.2
	CCT	94.99±0.0	82.03±0.3	88.27±0.2	73.97±0.1	85.04±0.1
	CPS	95.38±0.0	84.51±0.4	87.42±0.5	75.35±0.0	85.94±0.0
	U^{2}PL	95.44±0.0	85.97±0.3	85.73±0.5	75.21±0.1	85.85±0.0
	UniMatch	95.58±0.1	86.66±2.0	85.94±1.7	75.84±0.2	86.26±0.1
	SgeMind	95.40±0.1	83.98±0.6	88.35±0.5	75.60±0.1	86.10±0.1
	WSCL	95.70±0.0	**88.30±0.6**	84.54±0.2	76.02±0.3	86.38±0.2
	Semi-LM	**96.04±0.0**	87.52±0.2	88.02±0.3	**78.20±0.0**	**87.77±0.0**
14% (20)	SL	95.61±0.0	87.39±0.5	85.07±0.6	75.76±0.3	86.21±0.2
	MT	95.88±0.0	88.02±0.7	86.21±1.2	77.14±0.4	87.09±0.2
	CCT	95.79±0.0	86.58±0.6	87.50±1.3	77.04±0.5	87.03±0.3
	CPS	95.76±0.0	87.72±0.0	85.73±0.0	76.54±0.0	86.71±0.0
	U^{2}PL	95.88±0.0	**88.67±0.2**	85.40±0.2	76.99±0.0	87.00±0.0
	UniMatch	95.81±0.0	87.05±1.2	87.04±1.5	77.03±0.3	87.02±0.2
	SgeMind	95.65±0.0	84.21±0.8	89.98±1.0	76.97±0.1	86.99±0.1
	WSCL	95.89±0.1	86.88±1.4	87.83±2.0	77.51±1.0	87.33±0.6
	Semi-LM	**96.17±0.0**	85.94±0.6	**91.21±0.5**	**79.37±0.1**	**88.50±0.0**
100%	SL	95.79±0.1	85.99±0.3	88.29±0.5	77.19±0.6	87.13±0.4

*** Supervised Learning (SL), Mean Teacher (MT).**

4) Results on the Luding Dataset: Table VI shows that Semi-LM performs best in IoU and F_1-Score across all partition protocols on the Luding dataset. Fig. 16 illustrates that, unlike other datasets, the Luding dataset, due to its high spatial resolution and dense vegetation coverage, exhibits significant landslide damage on the ground. This reduces the difficulty of LM. Consequently, the gap between the SL baseline and SSL narrows, but Semi-LM still achieves the highest OA, IoU, and F_1-Score. Notably, under the 3% partition protocol, Semi-LM does not achieve the highest Precision and Recall but demonstrates a better balance between precision and recall through the highest OA. Quantitatively, the recall of the SL baseline is only 0.57% higher than that of Semi-LM, while its precision is 9.24% lower, the lowest among all methods. Similarly, WSCL has a precision of only 0.78% higher than Semi-LM, but its recall is 3.48% lower, also the lowest among all methods.

5) Results on the LRSTTC Dataset: Table VII reports the performance of the SL baseline and other SSL methods along with Semi-LM on the LRSTTC dataset. Specifically, under the 5%, 20%, and 30% partition protocols, Semi-LM improves the F_1-Score by 8.5%, 5.77%, and 5.67% over the SL baseline, respectively. Compared to UniMatch that the second-best method, Semi-LM also increases the F_1-Score from 53.52% to 57.86% under the 5% partition protocol. Additionally, the teacher–student framework of Semi-LM ensures training stability and produces more consistent results, with a standard deviation lower than that of other methods without teacher–student framework. Fig. 17 shows partial visualization results on the LRSTTC dataset. It can be observed that in mountainous canyon areas, numerous interfering features with spectra similar to landslides cause a higher number of false positives for local methods. However, Semi-LM, with its strong capability to learn from unlabeled data, achieves the most accurate LM results.

D. Ablation Studies

1) MCL: To investigate the impact of different perturbation perspectives on LM, we conduct an ablation experiment on the Bijie dataset under the 2% partition protocol. Table VIII shows that F0-M0-I0 achieves the lowest scores across all metrics. Compared to the single perturbation perspective F1-M0-I0, F0-M1-I0, and F0-M0-I1, the F_1-Score of F0-M0-I0 decreases by 16.11%, 18.11%, and 20.56%, respectively. This indicates that different perturbation perspectives each have a positive effect on LM. This is because each perturbation perspective allows the model to learn some general, common, and robust features from pseudo-labeled data and avoids overfitting to labeled data. Moreover, input perturbation provides additional information to decouple confirmation bias. Notably, reliable LM gains do not linearly increase with additional input perturbations. Specifically, F0-M0-I1 increases the IoU and F_1-Score by 21.34% and 20.56%, respectively, compared to F0-M0-I0. Adding another input perturbation further improves IoU and F_1-Score by 2.44% and 2%, respectively. However, when the number of input perturbations is increased to 3, IoU and F_1-Score only increase by 0.59% and 0.47%, respectively, while the computational burden increases significantly. Thus, the number of input perturbations is determined to be 2. This also verifies the effectiveness and rationality of the two feed-forward perturbations in MCL. Fig. 18 shows partial high-level feature activation maps of Semi-LM based on Grad-CAM. The

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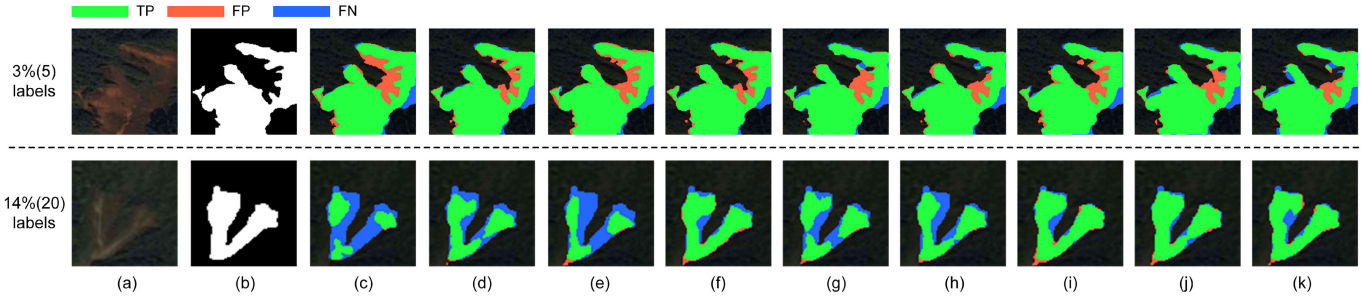


Fig. 16. LM visual comparison of Semi-LM with SOTA methods on the Luding dataset. (a) Input RSLIs. (b) Label. (c) SL. (d) MT. (e) CCT. (f) CPS. (g) U^{2}PL. (h) UniMatch. (i) SegMind. (j) WSCL. (k) Semi-LM.

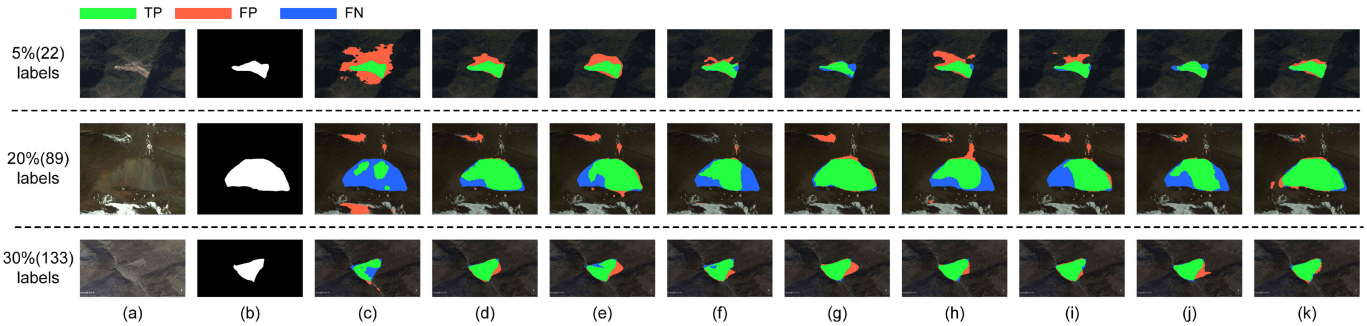


Fig. 17. LM visual comparison of Semi-LM with SOTA methods on the LRSTTC dataset. (a) Input RSLIs. (b) Label. (c) SL. (d) MT. (e) CCT. (f) CPS. (g) U^{2}PL. (h) UniMatch. (i) SegMind. (j) WSCL. (k) Semi-LM.

TABLE VII

COMPARISONS BETWEEN SEMI-LM WITH SOTA METHODS ON THE LRSTTC DATASET

Partition Protocol	Methods	OA /%	Precision /%	Recall /%	IoU /%	F_1-Score /%
5% (22)	SL	88.82±0.8	42.53±2.4	59.17±2.6	32.78±0.7	49.36±0.8
	MT	89.57±0.2	45.24±0.8	63.98±0.1	36.06±0.4	53.00±0.5
	CCT	90.66±0.5	49.42±2.5	54.73±1.6	35.05±1.3	51.89±1.4
	CPS	92.23±0.1	59.62±1.1	48.03±1.9	36.21±0.9	53.16±0.9
	U^{2}PL	91.09±0.2	51.66±1.4	50.79±2.0	34.38±0.2	51.16±0.3
	UniMatch	91.65±0.3	54.96±2.2	52.37±2.6	36.54±0.4	53.52±0.4
	SgeMind	92.54±0.2	**63.18±2.6**	45.61±2.3	35.94±0.7	52.88±0.7
	WSCL	91.71±0.1	55.57±1.3	49.42±0.9	35.40±0.2	52.29±0.2
	Semi-LM	**91.97±0.1**	55.87±0.8	**60.02±0.6**	**40.71±0.2**	**57.86±0.2**
20% (89)	SL	91.16±0.1	51.66±0.6	**60.07±0.9**	38.45±0.4	55.54±0.5
	MT	92.88±0.0	62.21±0.6	57.40±1.4	42.54±0.6	59.69±0.6
	CCT	92.40±0.2	59.17±2.3	56.67±3.1	40.64±1.0	57.78±1.0
	CPS	92.75±0.1	62.15±0.6	54.04±0.4	40.66±0.5	57.81±0.5
	U^{2}PL	92.56±0.1	60.13±0.9	56.54±0.2	41.12±0.4	58.28±0.4
	UniMatch	92.72±0.7	63.11±6.9	54.08±3.7	40.65±1.2	57.79±1.2
	SgeMind	93.28±0.2	67.56±2.1	51.90±2.0	41.52±1.4	58.66±1.4
	WSCL	**93.51±0.1**	**69.07±1.2**	53.21±0.1	42.97±0.4	60.11±0.4
	Semi-LM	93.48±0.0	67.44±0.5	56.20±0.3	**44.21±0.3**	**61.31±0.3**
30% (133)	SL	91.71±0.5	54.50±3.2	62.61±2.7	41.00±1.1	58.14±1.1
	MT	93.16±0.1	65.72±1.9	53.75±1.8	41.92±0.3	59.08±0.3
	CCT	92.73±0.7	62.10±5.3	56.57±3.3	41.76±1.2	58.90±1.2
	CPS	92.69±0.0	60.16±0.7	**60.77±1.1**	43.32±0.1	60.45±0.1
	U^{2}PL	92.76±0.1	61.29±1.0	57.55±0.6	42.21±0.7	59.36±0.7
	UniMatch	93.25±0.1	66.05±0.8	54.69±1.3	42.68±1.0	59.83±0.9
	SgeMind	93.48±0.1	68.28±1.2	54.40±1.6	43.40±0.8	60.53±0.8
	WSCL	93.04±0.2	62.74±2.0	60.20±3.1	44.27±1.6	61.36±1.5
	Semi-LM	**93.79±0.1**	**68.80±1.8**	59.58±1.7	**46.85±0.2**	**63.81±0.2**
100%	SL	93.70±0.2	66.70±2.2	63.06±1.7	47.89±0.3	64.77±0.2

*** Supervised Learning (SL), Mean Teacher (MT).**

figure shows that F0-M0-I0 can only focus on some prominent landslide features and fails to differentiate between similar

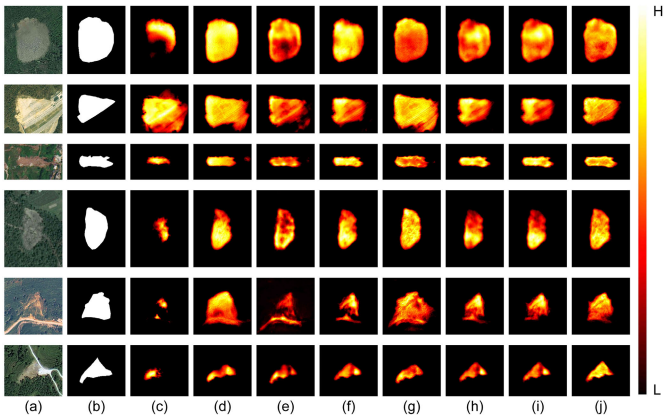


Fig. 18. Typical class activation maps results of LM on the Bijie dataset under different perturbation perspectives. (a) Input RSLIs. (b) Label. (c) F0-M0-I0. (d) F1-M0-I0. (e) F0-M1-I0. (f) F0-M0-I1. (g) F1-M1-I0. (h) F0-M0-I2. (i) F0-M0-I3. (j) F1-M2-I2.

interfering objects, resulting in poorer LM results. In contrast, after incorporating different perspectives consistency learning, the model learns more feature representations, which enhances its ability to focus on landslides.

2) Assign Stage of Feature Perturbation: To validate the proposed FPA strategy for LM, we apply the same feature perturbation assignment to the five stages of the U-Net, as illustrated in Fig. 11. The performance metrics under different stages of feature perturbation assignment in the Bijie and Luding datasets are shown in Fig. 19. The figure highlights the specific evaluation values for the best- and worst-performing perturbation assignment stages. From the figure, it is evident that the influence of perturbation assignment stages on

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TABLE VIII
COMPARISONS BETWEEN DIFFERENT PERTURBATION PERSPECTIVES ON THE BIJIE DATASET

Serial Name	Perturbation Perspectives			OA /%	Precision /%	Recall /%	IoU /%	F_1-Score /%
	Feature	Model	Input					
F0-M0-I0				88.19±0.2	66.32±1.3	40.83±0.8	33.82±0.8	50.54±0.9
F1-M0-I0	✓			90.54±0.1	69.63±1.7	64.00±1.7	49.98±0.3	66.65±0.2
F0-M1-I0		✓		90.68±0.0	68.26±0.3	69.04±0.8	52.26±0.3	68.65±0.3
F0-M0-I1			✓	92.57±0.0	**83.61±0.3**	61.85±0.5	55.16±0.3	71.10±0.3
F1-M1-I0	✓	✓		90.97±0.1	70.09±0.8	67.94±0.8	52.66±0.2	68.99±0.2
F0-M0-I2			✓✓	92.91±0.0	83.09±0.7	65.25±0.0	57.60±0.2	73.10±0.2
F0-M0-I3			✓✓✓	92.95±0.1	82.48±0.7	66.39±0.3	58.19±0.5	73.57±0.4
F1-M1-I2	✓	✓	✓✓	**93.28±0.0**	79.71±0.3	**73.13±0.2**	**61.65±0.2**	**76.28±0.2**

***F represents feature perturbation, M represents model perturbation, and I represents input perturbation.**

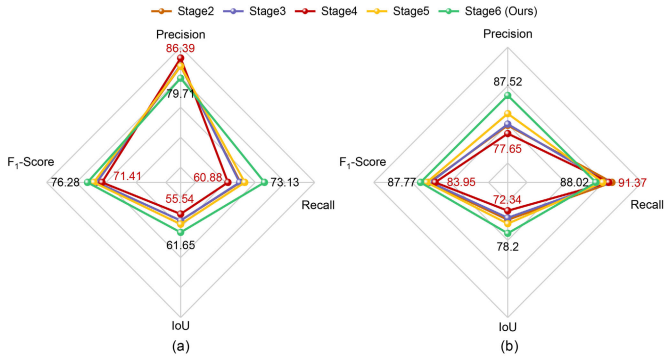


Fig. 19. Performance metrics under different assigned stages of feature perturbation. (a) 2% partition protocol of the Bijie dataset. (b) 3% partition protocol of the Luding dataset.

accuracy follows a similar pattern across different datasets. Under the 2% partition protocol of Bijie dataset, assigned perturbation at Stage 6 yields the best performance, with IoU and F_1-Score increasing by 6.11% and 4.87%, respectively, compared to the worst-performing Stage 4. Similarly, under the 3% partition protocol of the Luding dataset, Stage 6 achieves the IoU and F_1-Score improvements of 5.86% and 3.82%, respectively, over Stage 4. The results demonstrate that feature perturbation assignment at decoding stages consistently outperforms assignments at encoding stages. This finding aligns with the cluster assumption visualized in Fig. 11, which shows that the LM performance is optimized when feature perturbation is assigned to Stage 6. At this stage, the low-density regions align most closely with decision boundaries, thereby enhancing the ability of model to capture and classify landslide features effectively.

3) Dual-Branch Anchor CutMix: To demonstrate that the proposed DACM can reduce the disruption of both shallow and deep features and microtopographic information for landslide targets by generating a NO-Cut zone. We compare DACM with the original CutMix through experimental evaluation. Table IX shows that on the Bijie and LRSTTC datasets, Semi-LM and UniMatch equipped with DACM achieved superior performance metrics compared to those equipped with CutMix. In the Bijie dataset, the largest improvement of Semi-LM occurs under the 2% partition protocol, where DACM enhances the IoU and F_1-Score by 2.28% and 1.78%, respectively, compared to CutMix. In the LRSTTC dataset, the greatest improvement of Semi-LM is observed under the 30% partition protocol, with DACM increasing IoU and F_1-Score by 2.78% and 2.64%, respectively, over CutMix. Compared to Semi-LM, the accuracy improvement of UniMatch equipped with DACM is more significant. In the Bijie dataset, UniMatch equipped with DACM outperforms CutMix's F_1-Score by more than 2% in all cases, while IoU achieves a maximum improvement of 3.14% under the 30% partition protocol. In the LRSTTC dataset under the 5% partition protocol, DACM resulted in a maximum improvement of 1.65% in IoU and 1.74% in F_1-Score. Although CutMix achieves higher precision in most partition protocols, it results in a higher number of false negatives. Specifically, Semi-LM equipped with CutMix has an average of 1.46% higher than DACM in precision, but its recall is 3.91% lower on average. A similar pattern can be observed for UniMatch. Under the 6%, 20%, and 30% partition protocols in Bijie, and the 30% partition protocol in LRSTTC, CutMix achieves higher precision than DACM, but its recall is significantly lower, resulting in a large number of missed landslides. Consequently, DACM effectively improves the LM performance and detects more landslides.

E. Discussion

1) Basic Model: To verify that Semi-LM remains effective with different basic models, we conduct experiments using ResNet50-based DeepLabV3Plus, with results presented in Table X. Similar to U-Net, Semi-LM with ResNet50-based DeepLabV3Plus achieves the best IoU and F_1-Score across five datasets. Specifically, in the Bijie dataset, Semi-LM achieves the highest performance metrics in all indicators except precision. Compared to the SL baseline, Semi-LM improves IoU and F_1-Score by 12.13% and 10.13%, respectively. When compared to the second-best SSL method, the IoU and F_1-Score of Semi-LM are higher by 3.65% and 2.88%, respectively. In the Sichuan, Nepal, Luding, and LRSTTC datasets, the F_1-Score of Semi-LM exceeds that of the second-best SSL method by 1.62%, 2.4%, 0.74%, and 1.87%, respectively. It is noteworthy that although Semi-LM does not always achieve the highest precision or recall in some datasets, it avoids extreme values in individual metrics, demonstrating a better balance between precision and recall. Additionally, the ranking of the same comparative methods varies across different datasets, indicating that these methods have weaker handling capabilities for different styles of data

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TABLE IX
COMPARISONS BETWEEN DACM WITH CUTMIX METHODS ON THE BIJIE AND LRSTTC DATASETS

Dataset	Partition Protocol	Methods	OA /%	Precision /%	Recall/ %	IoU/ %	F_1-Score /%	OA /%	Precision /%	Recall /%	IoU /%	F_1-Score /%
			Semi-LM					UniMatch				
Bijie	2% (10)	CutMix	93.05±0.0	**81.36±0.3**	68.71±0.1	59.37±0.1	74.50±0.0	92.62±0.0	81.07±1.5	65.38±1.9	56.67±0.7	72.34±0.6
		DACM	**93.28±0.0**	79.71±0.3	**73.13±0.2**	**61.65±0.2**	**76.28±0.2**	**93.10±0.1**	**81.94±1.6**	**68.46±2.5**	**59.42±1.0**	**74.54±0.8**
	6% (30)	CutMix	93.43±0.0	**80.45±1.1**	73.44±1.0	62.29±0.1	76.77±0.1	93.25±0.5	**84.80±2.9**	66.27±0.8	59.25±2.0	74.39±1.6
		DACM	**93.49±0.0**	79.25±1.0	**75.79±0.9**	**63.23±0.0**	**77.47±0.0**	**93.69±0.1**	84.23±0.1	**70.46±0.8**	**62.25±0.7**	**76.73±0.5**
	20% (104)	CutMix	94.73±0.0	**86.57±0.5**	76.18±0.5	68.12±0.1	81.03±0.1	94.58±0.1	**89.90±1.6**	71.42±2.0	66.07±1.0	79.56±0.7
		DACM	**94.96±0.0**	85.20±0.4	**79.79±0.1**	**70.08±0.2**	**82.41±0.1**	**94.97±0.2**	88.61±1.4	**75.75±0.8**	**69.01±0.6**	**81.67±0.4**
	30% (154)	CutMix	95.65±0.0	**89.82±0.1**	79.60±0.2	73.01±0.1	84.40±0.1	95.12±0.0	**90.49±0.7**	74.83±1.2	69.36±0.6	81.91±0.4
		DACM	**95.82±0.0**	88.08±0.3	**82.94±0.4**	**74.57±0.1**	**85.43±0.1**	**95.56±0.3**	89.52±1.4	**79.29±2.8**	**72.50±1.7**	**84.05±1.2**
LRSTTC	5% (22)	CutMix	91.85±0.0	55.49±0.6	57.27±0.9	39.23±0.2	56.35±0.2	91.65±0.3	54.96±2.2	52.37±2.6	36.54±0.4	53.52±0.4
		DACM	**91.97±0.1**	**55.87±0.8**	**60.02±0.6**	**40.71±0.2**	**57.86±0.2**	**91.83±0.1**	**55.59±0.3**	**55.00±2.5**	**38.19±1.2**	**55.26±1.2**
	20% (89)	CutMix	**93.56±0.0**	**71.02±0.1**	50.44±0.1	41.83±0.1	58.99±0.1	92.72±0.7	63.11±6.9	54.08±3.7	40.65±1.2	57.79±1.2
		DACM	93.48±0.0	67.44±0.5	**56.20±0.3**	**44.21±0.3**	**61.31±0.3**	**93.18±0.2**	**65.54±1.6**	**54.48±1.0**	**42.32±0.2**	**59.47±0.2**
	30% (133)	CutMix	93.65±0.0	**69.82±1.2**	54.46±1.0	44.07±0.1	61.17±0.1	**93.25±0.1**	**66.05±0.8**	54.69±1.3	42.68±1.0	59.83±0.9
		DACM	**93.79±0.1**	68.80±1.8	**59.58±1.7**	**46.85±0.2**	**63.81±0.2**	92.99±0.2	62.59±2.1	**59.55±2.1**	**43.86±0.9**	**60.98±0.8**

TABLE X
COMPARISONS BETWEEN DEEPLABV3PLUS-BASED SEMI-LM WITH SOTA METHODS ON THE FIVE DATASETS

Dataset	Partition Protocol	Metrics /%	SL	MT	CCT	CPS	U^{2}PL	UniMatch	SegMind	WSCL	Semi-LM
Sichuan	7% (5)	OA	82.64±0.8	81.81±0.6	83.52±0.6	83.76±0.2	78.83±0.1	80.62±0.8	81.72±0.3	84.26±1.5	**84.41±0.6**
		Precision	59.34±2.0	56.13±1.3	61.57±1.2	60.93±0.3	50.79±0.1	53.70±1.4	56.38±0.9	**63.59±6.5**	61.89±2.1
		Recall	63.89±3.1	74.83±1.3	63.96±2.5	70.27±1.3	**80.26±0.0**	78.73±1.4	70.20±1.0	69.84±8.1	74.17±2.9
		IoU	44.43±2.1	47.19±0.6	45.73±1.9	48.44±0.7	45.15±0.1	46.88±1.1	45.47±0.1	49.00±0.8	**50.81±0.0**
		F_1-Score	61.49±2.0	64.12±0.5	62.73±1.8	65.26±0.7	62.21±0.1	63.83±1.0	62.51±0.1	65.77±0.7	**67.39±0.0**
Bijie	2% (10)	OA	89.27±0.3	91.68±0.1	91.59±0.0	91.96±0.0	92.15±0.0	93.07±0.1	91.08±0.2	92.29±0.0	**93.41±0.0**
		Precision	62.47±2.5	78.17±1.6	75.60±1.3	83.04±0.1	82.43±1.0	**86.11±1.7**	70.66±0.7	80.11±1.6	82.76±1.3
		Recall	70.01±6.5	60.70±1.4	63.75±2.5	57.28±0.2	59.58±0.6	63.34±0.2	67.80±1.1	63.69±1.8	**70.06±1.0**
		IoU	48.98±1.9	51.86±0.3	52.81±1.1	51.28±0.1	52.85±0.1	57.46±0.5	52.90±0.8	54.94±0.6	**61.11±0.1**
		F_1-Score	65.73±1.7	68.30±0.3	69.11±1.0	67.80±0.1	69.16±0.1	72.98±0.4	69.20±0.7	70.92±0.5	**75.86±0.1**
Nepal	15% (30)	OA	99.04±0.0	99.05±0.0	99.04±0.0	99.12±0.1	**99.21±0.0**	99.15±0.0	99.05±0.1	99.16±0.0	99.02±0.0
		Precision	42.79±4.7	42.86±1.4	42.90±2.7	50.83±9.7	**55.48±3.6**	49.83±4.1	43.39±3.4	50.63±3.5	42.76±0.4
		Recall	38.61±2.6	40.08±0.9	40.73±3.6	40.38±7.7	32.84±2.5	38.78±4.1	40.55±2.6	37.25±1.8	**49.51±0.9**
		IoU	25.24±0.9	26.13±0.9	26.28±1.2	27.75±2.0	25.87±1.1	27.57±0.7	26.35±0.6	27.20±0.2	**29.77±0.4**
		F_1-Score	40.30±1.2	41.42±1.1	41.60±1.5	43.41±2.5	41.09±1.4	43.22±0.9	41.71±0.7	42.77±0.2	**45.88±0.4**
Luding	3% (5)	OA	94.01±0.3	93.65±0.0	94.39±0.2	94.78±0.0	**95.14±0.0**	93.64±1.1	94.06±0.3	94.89±0.1	95.07±0.0
		Precision	81.63±1.2	75.70±0.1	79.08±0.8	83.06±0.4	**88.66±0.1**	75.79±4.9	78.41±1.2	82.72±0.5	82.39±0.2
		Recall	81.18±1.5	89.37±0.3	88.74±0.9	85.01±0.4	80.12±0.1	**89.99±1.4**	87.27±0.7	86.38±0.4	88.33±0.2
		IoU	68.64±1.5	69.45±0.3	71.87±1.1	72.45±0.1	72.66±0.1	69.78±3.5	70.36±0.9	73.18±0.4	**74.30±0.2**
		F_1-Score	81.39±1.1	81.97±0.2	83.62±0.7	84.03±0.0	84.17±0.0	82.15±2.4	82.60±0.6	84.51±0.3	**85.25±0.1**
LRSTTC	5% (22)	OA	88.64±0.6	89.66±0.2	90.32±0.5	91.96±0.4	90.34±0.2	92.02±0.3	90.63±0.3	92.25±0.1	**92.47±0.0**
		Precision	41.15±1.9	45.24±0.7	47.73±2.3	57.79±3.9	47.74±1.2	57.48±2.5	49.10±1.3	59.59±1.2	**60.40±0.5**
		Recall	53.73±1.8	**59.15±0.6**	52.35±3.7	48.92±2.4	51.28±3.4	51.39±1.1	50.02±1.3	48.87±2.9	52.35±0.2
		IoU	30.32±0.5	34.46±0.2	33.2±1.81	35.88±0.2	32.75±0.8	37.19±0.8	32.91±0.1	36.65±1.4	**38.97±0.2**
		F_1-Score	46.53±0.6	51.26±0.2	49.82±2.0	52.81±0.2	49.34±0.9	54.21±0.9	49.52±0.2	53.62±1.5	**56.08±0.2**

*** Supervised Learning (SL), Mean Teacher (MT).**

and do not generalize as well as Semi-LM. It can be observed that there is no significant difference in LM results between SSL methods designed for natural images and those designed for remote sensing images, but both outperform SL by a considerable margin. Additionally, by learning from unlabeled data more effectively, Semi-LM reduces the interference from nonlandslide features and identifies more subtle landslide characteristics, leading to more accurate landslide boundaries, as shown in Fig. 20.

2) Computational Load: We compare the computational load of SSL methods using ResNet50-based DeepLabV3Plus as the basic model across datasets with high (Bijie), medium (Nepal), and low (Sichuan) training data quantity, as shown in Fig. 21. The figure reveals that training time scales proportionally with the quantity of training data. Among the three datasets, U^{2}PL, CPS, and SegMind combine techniques such as pseudo-labeling, consistency learning, and contrastive learning, resulting in longer training times compared to other methods. In contrast, WSCL, which only includes input perturbation consistency learning, has the shortest training time. Semi-LM ranks in the middle in terms of training duration. Specifically, in the Bijie dataset, Semi-LM takes 23 min longer than WSCL, but it improves the F_1-Score by 4.94%.

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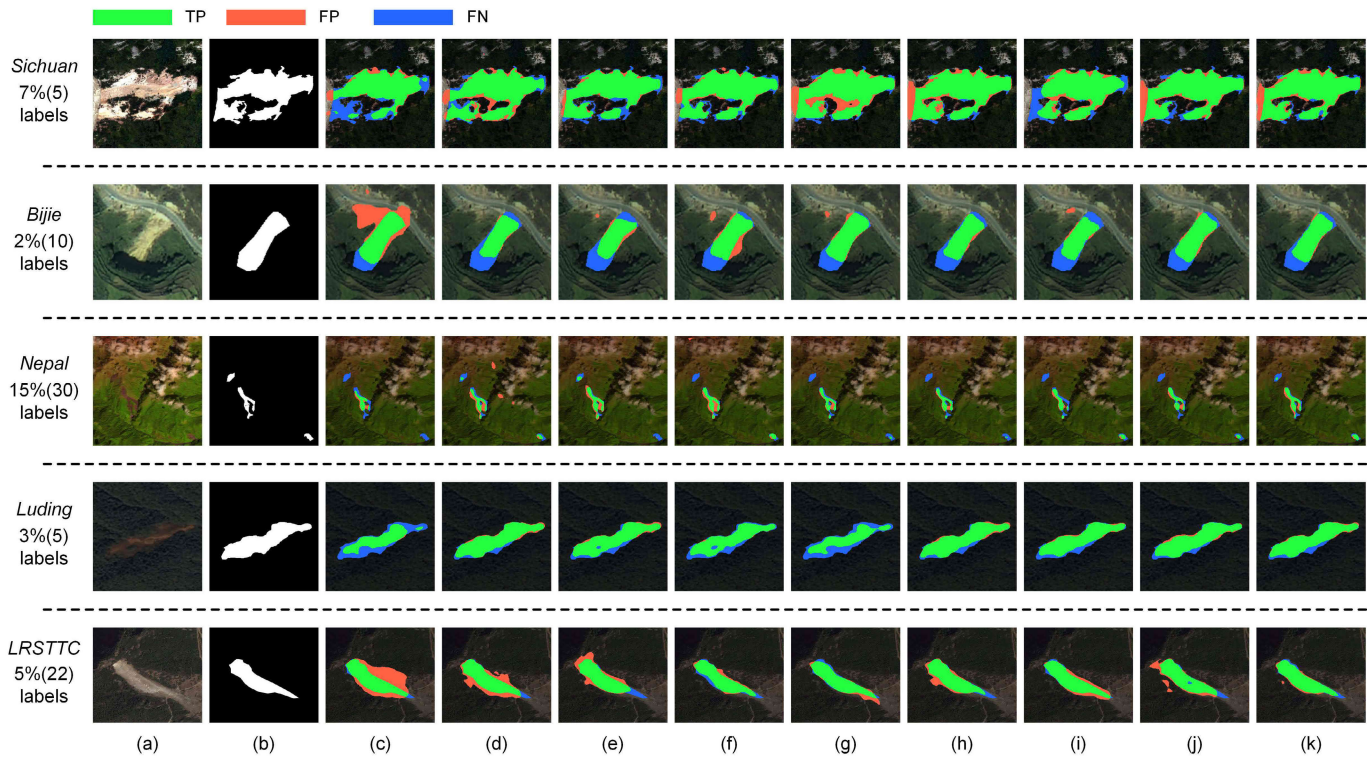


Fig. 20. LM visual comparison of DeepLabV3Plus-based Semi-LM with SOTA methods on the five datasets. (a) Input RSLIs. (b) Label. (c) SL. (d) MT. (e) CCT. (f) CPS. (g) U^{2}PL. (h) UniMatch. (i) SegMind. (j) WSCL. (k) Semi-LM.

TABLE XI
COMPARATIVE RESULTS OF TEACHER MODELS FROM U-NET-BASED SEMI-LM WITH DIFFERENT INTENSITIES OF NOISE

Dataset and Partition Protocol	Noise	OA /%	Precision /%	Recall /%	IoU /%	F_1-Score /%	Training Time/ min
Sichuan 7% (5)	None	85.43±0.7	64.49±2.3	**73.73±3.0**	52.35±1.2	68.72±1.0	3
	(-0.1,01)	**87.21±0.2**	**70.31±1.4**	71.25±1.6	**54.74±0.2**	**70.75±0.2**	3
	(-0.9,0.9)	86.42±0.6	68.12±1.9	70.56±1.2	53.02±0.9	69.29±0.8	3
Bijie 2% (10)	None	93.14±0.1	79.09±0.2	72.82±0.4	61.06±0.3	75.82±0.2	33
	(-0.1,01)	**93.28±0.0**	79.71±0.3	**73.13±0.2**	**61.65±0.2**	**76.28±0.2**	33
	(-0.9,0.9)	93.17±0.0	**80.53±0.1**	70.91±0.2	60.54±0.1	75.42±0.1	33
Nepal 15% (30)	None	99.22±0.0	53.97±0.6	48.56±0.3	34.34±0.3	51.12±0.3	22
	(-0.1,01)	**99.23±0.0**	**54.58±0.4**	**49.45±0.4**	**35.03±0.2**	**51.89±0.2**	22
	(-0.9,0.9)	99.22±0.0	54.13±0.8	48.13±1.4	34.17±0.5	50.93±0.6	22
Luding 14% (20)	None	95.93±0.1	86.85±0.5	88.17±0.4	77.79±0.5	87.51±0.3	15
	(-0.1,01)	**96.17±0.0**	85.94±0.6	**91.21±0.5**	**79.37±0.1**	**88.50±0.0**	15
	(-0.9,0.9)	95.92±0.1	**87.47±0.7**	87.23±0.9	77.53±0.4	87.34±0.3	15
LRSTTC 5% (22)	None	91.78±0.1	54.88±0.6	59.48±0.8	39.94±0.1	57.08±0.2	48
	(-0.1,01)	**91.97±0.1**	55.87±0.8	**60.02±0.6**	**40.71±0.2**	**57.86±0.2**	48
	(-0.9,0.9)	91.95±0.1	**56.06±0.7**	57.49±1.8	39.61±0.6	56.74±0.6	49

In the Nepal dataset, Semi-LM requires 14 more minutes than WSCL, but the F_1-Score increases from 42.77% to 45.88%. In the Sichuan dataset, the training time of Semi-LM is comparable to that of WSCL, while the F_1-Score improves by 1.62%. Thus, Semi-LM achieves a more effective LM by enhancing the ability of model to extract and learn features from unlabeled data, despite a partial decrease in computational efficiency.

3) Teacher Model Noise: We conduct experiments by feeding noise of varying intensities into the teacher model of the U-Net-based Semi-LM, and the results are presented in Table XI. In the table, "None" indicates that no noise is fed into the teacher model, while specific values represent the range of noise intensities. The experimental results indicate that when Noise(,) $\in (-0.1, 0.1)$, Semi-LM achieves higher scores on all metrics except recall compared to the case without noise feeding. This demonstrates that introducing Gaussian noise into the teacher model positively influences LM performance. However, as the noise intensity increases further, the OA, recall, IoU, and F_1-Score all decline. This deterioration likely occurs because excessive noise disrupts the original input information, causing the student model to learn less discriminative or even erroneous features during the training process. Additionally, an analysis of training time shows that noise feeding incurs virtually no additional computational cost.

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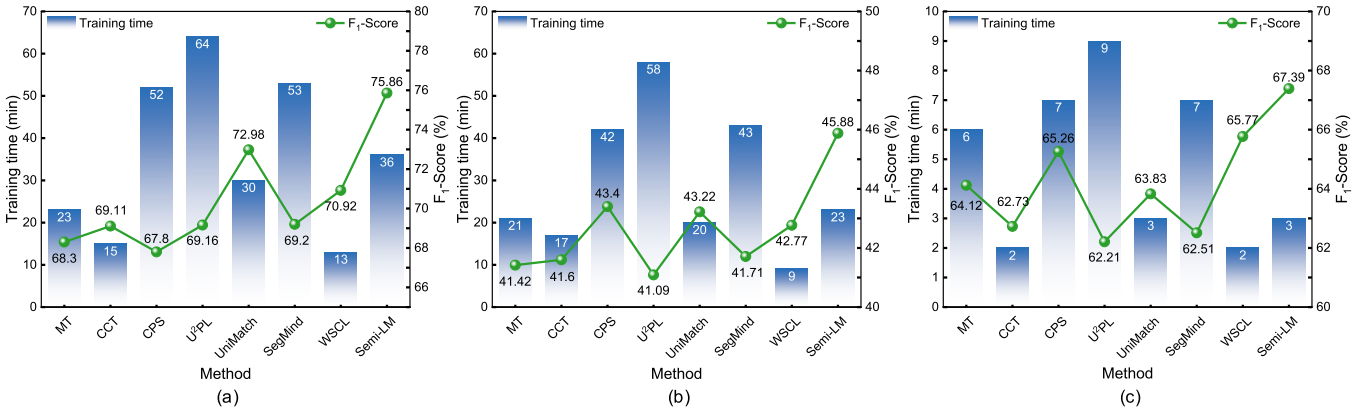


Fig. 21. Computational load comparisons between DeepLabV3Plus-based Semi-LM with SOTA methods on the three datasets. (a) 2% partition protocol of the Bijie dataset. (b) 15% partition protocol of the Nepal dataset. (c) 7% partition protocol of the Sichuan dataset.

V. Conclusion

To address the costly, labor-intensive, and time-consuming issue of pixel-level labeling of remote sensing images in the deep learning-based LM, we propose a reliable multiperspective consistency SSL framework, Semi-LM. This framework expands the perturbation space by integrating MCL into a unified objective function, enabling the learning of more discriminative features for unlabeled RSLIs. To prevent the degradation of both shallow and deep landslide features and microtopographic information in the original input data, we have designed a simple and effective strong perturbation-based augmentation method, DACM. Furthermore, we observe that the low-density regions of separated classes are positioned at the end of the decoding stage in the LM process, which helps determine the optimal landslide FPA stage. Semi-LM demonstrates superior performance across various partition protocols on five open-source landslide datasets, achieving a better balance between precision and recall and identifying more landslides. The visualization results and high-dimensional features indicate that the LM results of Semi-LM are closer to the ground truth, and multiperspective consistency is beneficial for LM. Ablation experiments on the basic model demonstrate that the proposed Semi-LM is a reliable and versatile framework. It can be combined with different basic models for LM, reducing the reliance on manual labeling. Finally, the discussion on computational load shows that Semi-LM achieves the highest performance with moderate training time. This provides new insights into reducing the labor and resource costs for other natural hazard mapping tasks.

In future work, we will conduct extensive cross-regional and cross-dataset LM experiments to further validate the generalizability and robustness of Semi-LM, providing a reliable solution for landslide monitoring in real-world scenarios. Additionally, combining the advantages of self-SL for LM will be a key focus of future work.

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